Probability: Concepts and Applications

Lecture notes, Lectures 1–10

Contents

Lecture 1. Combinatorial Analysis 4

1.1 The basic principle of counting 4

1.2 Permutations 4

1.3 Combinations 4

1.4 Multinomial coefficients 5

1.5 Integer solutions of equations 6

1.6 Selected exercises 7

Lecture 2. Axioms of Probability 9

2.1 Sample space and events 9

2.2 Axioms of probability 9

2.3 Equally likely outcomes 10

2.4 Probability as a continuous set function 11

2.5 Selected exercises 11

Lecture 3. Conditional Probability and Independence 14

3.1 Conditional probability 14

3.2 Total probability and Bayes's formula 14

3.3 Independent events 16

3.4 $P(\cdot \mid F)$ is a probability 17

3.5 Selected exercises 17

Lecture 4. Discrete Random Variables 20

4.1 Random variables and distribution functions 20

4.2 Expectation 20

4.3 Variance 21
4.4 Bernoulli and binomial random variables 21
4.5 The Poisson random variable 22
4.6 Other discrete distributions 23
4.7 Expected value of sums 23
4.8 Selected exercises 24

Lecture 5. Continuous Random Variables 26
5.1 Densities 26
5.2 Expectation and variance 26
5.3 The uniform distribution 27
5.4 Normal random variables 27
5.5 Exponential random variables 28
5.6 Other continuous distributions 28
5.7 The distribution of a function of a random variable 29
5.8 Selected exercises 29

Lecture 6. Jointly Distributed Random Variables 32
6.1 Joint distribution functions 32
6.2 Independent random variables 32
6.3 Sums of independent random variables 33
6.4 Conditional distributions 34
6.5 Order statistics 35
6.6 Functions of random variables: the Jacobian method 35
6.7 Exchangeable random variables 36
6.8 Selected exercises 36

Lecture 7. Properties of Expectation 39
7.1 Expectation of sums 39
7.2 Covariance and variance of sums 39
7.3 Conditional expectation 40

7.4 Conditional expectation and prediction **42**

7.5 Moment generating functions **42**

7.6 The multivariate normal distribution **43**

7.7 General definition of expectation **43**

7.8 Selected exercises **44**

Lecture 8. Limit Theorems **46**

8.1 Markov and Chebyshev inequalities; the weak law **46**

8.2 The central limit theorem **46**

8.3 The strong law of large numbers **47**

8.4 Other inequalities **48**

8.5 The Poisson approximation, quantified **48**

8.6 Selected exercises **49**

Lecture 9. Additional Topics in Probability **51**

9.1 The Poisson process **51**

9.2 Markov chains **52**

9.3 Surprise, uncertainty and entropy **53**

9.4 Coding theory and entropy **54**

9.5 Selected exercises **54**

Lecture 10. Simulation **56**

10.1 Random numbers and simulation **56**

10.2 Simulating continuous random variables **56**
10.2.1 The inverse transformation method 56
10.2.2 The rejection method 56

10.3 Simulating from discrete distributions **57**

10.4 Variance reduction **57**
10.4.1 Antithetic variables 57
10.4.2 Variance reduction by conditioning 58
10.4.3 Control variates 58

10.5 Selected exercises **58**

Lecture 1. Combinatorial Analysis

Probabilities on finite spaces with equally likely outcomes reduce to counting.

1.1 The basic principle of counting

Motivation. Equally likely probabilities are ratios of counts, and counting reduces to multiplying the choices at each stage.

Theorem 1.1 (Generalized counting principle)

If r experiments are performed in sequence so that the first has n_1 possible outcomes and, for each outcome of the first $i-1$ experiments, the i-th has n_i possible outcomes, then the r experiments together have $n_1 n_2 \cdots n_r$ possible outcomes.

Proof. For $r = 2$ list the outcomes as pairs (i, j), $1 \le i \le n_1$, $1 \le j \le n_2$: an $n_1 \times n_2$ array. Induct on r, treating the first $r-1$ experiments as one. ■

The number n_i may not depend on earlier outcomes, but the *set* of outcomes may.

Example 1.1 (License plates)

Plates of 3 letters and 4 digits: $26^3 \cdot 10^4 = 175{,}760{,}000$; with no repeats, $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 78{,}624{,}000$. Likewise there are 2^n functions from an n-set to $\{0, 1\}$.

1.2 Permutations

An ordered arrangement of n distinct objects is a **permutation**; there are $n! = n(n-1)\cdots 1$ of them, with $0! = 1$. The number of ordered selections of r objects out of n is $n(n-1)\cdots(n-r+1) = n!/(n-r)!$.

If the objects fall into r groups of $n_1, \ldots, n_r$ indistinguishable ones, each distinguishable arrangement corresponds to $n_1! \cdots n_r!$ labelled permutations, so there are

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}$$

distinguishable arrangements.

Example 1.2

(a) The letters of PEPPER (three P, two E, one R) give $6!/(3!\,2!\,1!) = 60$ arrangements.
(b) Four mathematics, three chemistry, two history and one language book, same subjects together: $4!\,(4!\,3!\,2!\,1!) = 6912$ arrangements.
(c) 6 men and 4 women ranked without ties: $10! = 3{,}628{,}800$; men and women ranked separately: $6!\,4! = 17{,}280$.

1.3 Combinations

The number of *unordered* groups of r objects chosen from n is

$$\binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{(n-r)!\,r!}, \qquad 0 \le r \le n,$$

since each group appears $r!$ times among ordered selections. Set $\binom{n}{r} = 0$ for $r < 0$ or $r > n$; $\binom{n}{r} = \binom{n}{n-r}$.

Proposition 1.2 (Pascal's identity)

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \text{ for } 1 \le r \le n.$$

Proof. Fix one object. The r-subsets containing it number $\binom{n-1}{r-1}$; those not containing it number $\binom{n-1}{r}$. ■

Theorem 1.3 (Binomial theorem)

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Proof. $(x_1+y_1)\cdots(x_n+y_n)$ is a sum of 2^n products, one per choice of x_i or y_i from each factor; those with k x's correspond to the $\binom{n}{k}$ k-subsets. Set $x_i = x$, $y_i = y$. ■

$x = y = 1$ gives $\sum_k \binom{n}{k} = 2^n$; $x = -1, y = 1$ gives $\sum_k (-1)^k \binom{n}{k} = 0$ $(n \ge 1)$.

Counting one set in two ways:

Proposition 1.4

$$\text{(a)} \sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}, \qquad \text{(b)} \binom{n+m}{r} = \sum_{k=0}^{r} \binom{n}{k}\binom{m}{r-k} \quad \text{(Vandermonde)}.$$

Proof. (a) Count (committee, chair) pairs from n people: committee of size k then chair, or chair (n ways) then any subset of the rest (2^{n-1}).

(b) Choose r people from n men and m women by first fixing the number k of men. ■

Example 1.3 (Antennas in a row)

n antennas in a line, m indistinguishable defectives. Orderings with no two defectives adjacent: the $n-m$ working ones create $n-m+1$ gaps, each taking at most one defective, so $\binom{n-m+1}{m}$. E.g. $n = 4, m = 2$: DWDW, DWWD, WDWD.

1.4 Multinomial coefficients

The number of ways to divide n distinct objects into r *labelled* groups of sizes $n_1, \ldots, n_r$ $(\sum n_i = n)$ is

$$\binom{n}{n_1, \ldots, n_r} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r} = \frac{n!}{n_1!\cdots n_r!},$$

the count of Section 1.2 (assigning group labels = arranging a word with n_i copies of label i).

Theorem 1.5 (Multinomial theorem)

$(x_1 + \cdots + x_r)^n = \sum_{n_1+\cdots+n_r=n} \binom{n}{n_1, \ldots, n_r} x_1^{n_1} \cdots x_r^{n_r}$, the sum over nonnegative integer vectors.

Example 1.4 (Labelled versus unlabelled groups)

Ten children into teams A and B of five: $\binom{10}{5,5} = 252$; unlabelled teams: 126.

Example 1.5 (Knockout tournament)

With $n = 2^m$ players, the first round has $n!/\{2^{n/2}(n/2)!\}$ pairings (arrange in a row, pair neighbours, forget orders), each with $2^{n/2}$ outcomes: $n!/(n/2)!$ outcomes. Round by round, the tournament has

$$\frac{n!}{(n/2)!} \cdot \frac{(n/2)!}{(n/4)!} \cdots \frac{2!}{1!} = n!$$

outcomes.

1.5 Integer solutions of equations

Motivation. Stars and bars turns placing identical objects into boxes into choosing separator positions.

Proposition 1.6 (Stars and bars)

There are $\binom{n-1}{r-1}$ vectors of *positive* integers with $x_1 + \cdots + x_r = n$, and $\binom{n+r-1}{r-1}$ vectors of *nonnegative* integers.

Proof. A positive solution chooses $r-1$ of the $n-1$ gaps between n stars as dividers. For nonnegative ones, $y_i = x_i + 1$ solves $\sum y_i = n + r$ positively. ■

So $\binom{n+r-1}{r-1}$ counts placements of n identical balls in r urns and the terms of $(x_1 + \cdots + x_r)^n$.

Example 1.6 (Investment)

\$20,000 in \$1000 units among 4 options: all invested, $\binom{23}{3} = 1771$; not necessarily all (a fifth "cash" variable), $\binom{24}{4} = 10{,}626$; each option at least one unit, $\binom{19}{3} = 969$.

Example 1.7 (Antennas revisited)

In Example 1.3, let $x_1, \ldots, x_{m+1}$ be the working antennas before, between and after the defectives: $\sum x_i = n - m$, $x_1, x_{m+1} \geq 0$, $x_2, \ldots, x_m \geq 1$. With $y_1 = x_1 + 1$, $y_{m+1} = x_{m+1} + 1$, a positive solution of $\sum y_i = n - m + 2$: again $\binom{n-m+1}{m}$. At least two working antennas between defectives: substitute $y_i = x_i - 1$ in the interior.

1.6 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): counting in two ways, and a proof of the multinomial theorem.

Exercise 1.11 (Classifying subsets by their largest element)

Show that for $1 \le k \le n$,

$$\sum_{i=k}^{n} \binom{i-1}{k-1} = \binom{n}{k}.$$

Solution.

Step 1. Let $\mathcal{A}$ be the k-subsets of $\{1,\dots,n\}$, $|\mathcal{A}| = \binom{n}{k}$. Each A has $\max A \ge k$, so $\mathcal{A}$ is the disjoint union of $\mathcal{A}_i = \{A : \max A = i\}$, $i = k,\dots,n$.

Step 2. $A \in \mathcal{A}_i$ is $\{i\}$ plus an arbitrary $(k-1)$-subset of $\{1,\dots,i-1\}$: $|\mathcal{A}_i| = \binom{i-1}{k-1}$.

Step 3. Sum over i. ■

Exercise 1.12 (Committees with a chair and a secretary)

Show that for $n \ge 1$,

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = 2^{n-2} n(n+1).$$

(The identity $\sum_k k\binom{n}{k} = n2^{n-1}$ is Proposition 1.4(a).)

Solution. Count triples (C, a, b): C a nonempty committee from n people, chair $a \in C$, secretary $b \in C$, $a = b$ allowed.

Step 1. (Committee first.) $\binom{n}{k}$ committees of size k, each with k^2 choices of (a,b): $\sum_{k=1}^{n} k^2\binom{n}{k}$.

Step 2. (Officers first, $a = b$.) n choices, the other $n-1$ people free: $n2^{n-1}$.

Step 3. (Officers first, $a \ne b$.) $n(n-1)$ pairs, the other $n-2$ free: $n(n-1)2^{n-2}$ (0 if $n = 1$).

Step 4. Steps 2 and 3 partition the set of Step 1:

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = n2^{n-1} + n(n-1)2^{n-2} = 2^{n-2}(2n + n^2 - n) = 2^{n-2} n(n+1).$$

■

Exercise 1.16 (Tournaments with ties)

A tournament of n contestants ends in an ordering with ties allowed (groups tied for first, next, and so on). Let $N(n)$ be the number of outcomes, $N(0) = 1$.
(a) Show that $N(n) = \sum_{i=1}^{n} \binom{n}{i} N(n-i)$ for $n \ge 1$.
(b) Show that this is equivalent to $N(n) = \sum_{i=0}^{n-1} \binom{n}{i} N(i)$.
(c) Find $N(3)$ and $N(4)$.

Solution.

(a) Classify by the number $i \in \{1,\dots,n\}$ tied for last. Such an outcome is determined by the set L of last players and an arbitrary outcome of the other $n-i$ (one if $i = n$, matching $N(0) = 1$), and conversely; so there are $\binom{n}{i} N(n-i)$ of them. Sum over i.

(b) Substitute $j = n - i$ and use $\binom{n}{n-j} = \binom{n}{j}$.

(c) From (b), $N(1) = N(0) = 1$, $N(2) = N(0) + 2N(1) = 3$,

$$N(3) = N(0) + 3N(1) + 3N(2) = 1 + 3 + 9 = 13,$$
$$N(4) = N(0) + 4N(1) + 6N(2) + 4N(3) = 1 + 4 + 18 + 52 = 75.$$

■

Exercise 1.19 (The multinomial theorem)

Prove Theorem 1.5.

Solution.

Step 1. Expanding the product of n factors gives the sum over words $w = (i_1, \dots, i_n) \in \{1, \dots, r\}^n$ of $x_{i_1} \cdots x_{i_n}$, the word recording the summand taken from each factor.

Step 2. If $n_j = \#\{t : i_t = j\}$, then $x_{i_1} \cdots x_{i_n} = x_1^{n_1} \cdots x_r^{n_r}$; grouping words by $(n_1, \dots, n_r)$,

$$(x_1 + \cdots + x_r)^n = \sum_{n_1 + \cdots + n_r = n} c(n_1, \dots, n_r)\, x_1^{n_1} \cdots x_r^{n_r},$$

with $c(n_1, \dots, n_r)$ the number of words with n_j copies of letter j.

Step 3. By Section 1.2, $c(n_1, \dots, n_r) = n!/(n_1! \cdots n_r!) = \binom{n}{n_1, \dots, n_r}$. ■

Lecture 2. Axioms of Probability

2.1 Sample space and events

The **sample space** S is the set of outcomes; an **event** is a subset. We use $E \cup F$, $EF = E \cap F$, $E^c = S \setminus E$, $\emptyset$; E, F are **mutually exclusive** if $EF = \emptyset$. Union and intersection are commutative, associative and mutually distributive, and

$$\Big(\bigcup_{i=1}^{n} E_i\Big)^c = \bigcap_{i=1}^{n} E_i^c, \qquad \Big(\bigcap_{i=1}^{n} E_i\Big)^c = \bigcup_{i=1}^{n} E_i^c \qquad \text{(De Morgan)}.$$

(An outcome lies outside every E_i iff it lies outside their union; apply this to the E_i^c for the second.)

2.2 Axioms of probability

Relative frequency $\lim n(E)/n$ motivates but does not define probability; we impose axioms.

Definition (Probability)

A function P on events is a **probability** if (i) $0 \le P(E) \le 1$;
(ii) $P(S) = 1$;
(iii) for mutually exclusive $E_1, E_2, \ldots$, $P\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} P(E_i)$.

$E_1 = S$, $E_i = \emptyset$ $(i \ge 2)$ forces $P(\emptyset) = 0$, and (iii) then gives finite additivity. (For uncountable S, P is defined on a suitable class of events.)

Proposition 2.1

(a) $P(E^c) = 1 - P(E)$.
(b) $E \subset F \Rightarrow P(E) \le P(F)$.
(c) $P(E \cup F) = P(E) + P(F) - P(EF)$.

Proof. (a) E and E^c are disjoint with union S.

(b) $F = E \cup E^c F$, a disjoint union.

(c) $E \cup F = E \cup E^c F$ and $F = EF \cup E^c F$, both disjoint; subtract. ∎

Theorem 2.2 (Inclusion–exclusion)

$$P\Big(\bigcup_{i=1}^{n} E_i\Big) = \sum_i P(E_i) - \sum_{i<j} P(E_i E_j) + \ldots + (-1)^{r+1} \sum_{i_1 < \cdots < i_r} P(E_{i_1} \cdots E_{i_r}) + \ldots + (-1)^{n+1} P(E_1 \cdots E_n).$$

Proof. For countable S, check each outcome contributes equally to both sides: one in no E_i contributes 0; one in exactly $m \ge 1$ contributes once on the left and $\binom{m}{1} - \binom{m}{2} + \cdots \pm \binom{m}{m} = 1 - \sum_{i=0}^{m} (-1)^i \binom{m}{i} = 1$ times on the right. (General S: induct with Proposition 2.1(c).) ∎

Truncations alternate between upper and lower bounds (Bonferroni):

$$P\Big(\bigcup E_i\Big) \le \sum P(E_i), \qquad P\Big(\bigcup E_i\Big) \ge \sum P(E_i) - \sum_{i<j} P(E_iE_j), \quad \dots$$

2.3 Equally likely outcomes

If $S = \{1, \dots, N\}$ with equal probabilities, $P(E) = |E|/N$. Treating objects as distinguishable never changes the answer and makes outcomes equally likely.

Example 2.1

(a) Three balls from 6 white and 5 black: $P(\text{one white, two black}) = \binom{6}{1}\binom{5}{2}/\binom{11}{3} = 4/11$; ordered draws give $3 \cdot 6 \cdot 5 \cdot 4/(11 \cdot 10 \cdot 9) = 4/11$.
(b) k of n balls drawn, one special: by symmetry $P(\text{special drawn}) = k/n$; directly $\binom{n-1}{k-1}/\binom{n}{k} = k/n$.
(c) Poker full house (rank and suits of the triple, then of the pair): $13\binom{4}{3} \cdot 12\binom{4}{2}/\binom{52}{5} = 3744/2{,}598{,}960 \approx 0.00144$.

Example 2.2 (Birthday problem)

With all 365^n assignments equally likely, n people have distinct birthdays with probability $365 \cdot 364 \cdots (366-n)/365^n$; a shared birthday has probability 0.507 at $n = 23$ and 0.970 at $n = 50$. (23 people form 253 pairs, each matching with probability 1/365.)

Example 2.3 (The card after the first ace)

A shuffled deck is turned up. Is the card after the first ace more likely the ace of spades or the two of clubs? Remove the ace of spades, order the other 51 cards (51! ways) and reinsert it right after the first ace: probability $51!/52! = 1/52$. The same holds for the two of clubs.

Example 2.4 (The matching problem)

N people pick hats at random; E_i: person i gets their own. $P(E_{i_1} \cdots E_{i_r}) = (N-r)!/N!$ for each of $\binom{N}{r}$ index sets, so

$$P\Big(\bigcup_i E_i\Big) = \sum_{r=1}^{N} (-1)^{r+1} \binom{N}{r} \frac{(N-r)!}{N!} = \sum_{r=1}^{N} \frac{(-1)^{r+1}}{r!},$$

and

$$P(\text{no match}) = \sum_{r=0}^{N} \frac{(-1)^r}{r!} \xrightarrow[N\to\infty]{} e^{-1} \approx 0.368.$$

Exactly k matches (choose them, no match among the other $N-k$):

$$P(\text{exactly } k) = \binom{N}{k} \frac{(N-k)!}{N!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} = \frac{1}{k!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} \approx \frac{e^{-1}}{k!},$$

a Poisson(1) probability (Lecture 4).

Example 2.5 (Couples at a round table)

Ten couples at a round table (19! circular orders); E_i: couple i together. Gluing r couples leaves $20-r$ units in $(19-r)!$ circular orders, each block in 2 ways:

$$P(\text{no couple together}) = \sum_{r=0}^{10} (-1)^r \binom{10}{r} \frac{2^r (19-r)!}{19!} \approx 0.3395.$$

2.4 Probability as a continuous set function

Motivation. Countable additivity makes probability respect monotone limits such as "eventually".

A sequence of events is **increasing** if $E_n \subset E_{n+1}$, with limit $\bigcup E_n$, and **decreasing** if $E_n \supset E_{n+1}$, with limit $\bigcap E_n$.

Proposition 2.3 (Continuity)

For an increasing or decreasing sequence, $\lim_n P(E_n) = P(\lim_n E_n)$.

Proof. Increasing: $F_1 = E_1$, $F_n = E_n E_{n-1}^c$ are disjoint with $\bigcup_1^n F_i = E_n$, $\bigcup_1^\infty F_i = \bigcup_1^\infty E_i$, so by (iii)

$$P\Big(\bigcup_i E_i\Big) = \sum_{i=1}^{\infty} P(F_i) = \lim_n \sum_{i=1}^{n} P(F_i) = \lim_n P(E_n).$$

Decreasing: apply this to E_n^c and use De Morgan. ■

Example 2.6

A fair coin is tossed forever. E_n (first n tosses tails) decreases to T (no head ever), so $P(T) = \lim 2^{-n} = 0$: a head appears with probability 1, though "all tails" is possible.

Coherent subjective probabilities obey the same axioms.

2.5 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): inclusion–exclusion by induction, Bonferroni, and two counting facts.

Exercise 2.8 (Counting partitions)

Let T_n be the number of partitions of $\{1, \dots, n\}$ into nonempty blocks, and put $T_0 = 1$.
(a) Show that $T_3 = 5$.
(b) Show that $T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k$ for $n \geq 1$, and deduce $T_4 = 15$.

Solution.

(a) The partitions of $\{1,2,3\}$ are $\{123\}$, $\{1\}\{23\}$, $\{2\}\{13\}$, $\{3\}\{12\}$ and $\{1\}\{2\}\{3\}$.

(b)

Step 1. A partition π of $\{1, \dots, n+1\}$ determines the block $B \ni n+1$ and the partition π' of $R = \{1, \dots, n\} \setminus B$ by the other blocks (empty if $R = \emptyset$); conversely $A \subseteq \{1, \dots, n\}$ and a partition π' of the complement give $\{A \cup \{n+1\}\} \cup \pi'$. These are inverse maps.

Step 2. With $k = |R|$: $\binom{n}{k}$ choices of A, then T_k choices of π' ($k \geq 1$) or one ($k = 0$):

$$T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k.$$

Step 3. $T_4 = 1 + 3T_1 + 3T_2 + T_3 = 1 + 3 + 6 + 5 = 15$ (then $T_5 = 52, \ldots, T_{10} = 115{,}975$). ■

Exercise 2.14 (Inclusion–exclusion by induction)

Prove Theorem 2.2 for an arbitrary sample space by induction on n.

Solution. With $E_I = \bigcap_{i \in I} E_i$, the claim is $P(\bigcup_{i=1}^n E_i) = \sum_I (-1)^{|I|+1} P(E_I)$ over nonempty $I \subseteq \{1, \ldots, n\}$.

Step 1. $n = 1$ is trivial; $n = 2$ is Proposition 2.1(c).

Step 2. Assume the claim for n events; let $U = \bigcup_{i=1}^n E_i$. By Proposition 2.1(c) and $UE_{n+1} = \bigcup_{i=1}^n (E_i E_{n+1})$,

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = P(U) + P(E_{n+1}) - P\Big(\bigcup_{i=1}^{n} E_i E_{n+1}\Big).$$

Step 3. Apply the hypothesis to $E_1, \ldots, E_n$ and to $E_1 E_{n+1}, \ldots, E_n E_{n+1}$, whose intersection over J is $E_{J \cup \{n+1\}}$:

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = \sum_J (-1)^{|J|+1} P(E_J) + P(E_{\{n+1\}}) - \sum_J (-1)^{|J|+1} P(E_{J \cup \{n+1\}}),$$

both sums over nonempty $J \subseteq \{1, \ldots, n\}$.

Step 4. Each nonempty $I \subseteq \{1, \ldots, n+1\}$ occurs once: $n+1 \notin I$ in the first sum, sign $(-1)^{|I|+1}$; $I = \{n+1\}$ as the middle term, sign $+1$; $I = J \cup \{n+1\}$ in the last sum, sign $-(-1)^{|J|+1} = (-1)^{|I|+1}$. ■

Exercise 2.16 (Bonferroni's inequality)

Show that for any events $E_1, \ldots, E_n$,

$$P(E_1 E_2 \cdots E_n) \geq P(E_1) + \cdots + P(E_n) - (n-1).$$

Solution.

Step 1. ($n = 2$.) $P(EF) = P(E) + P(F) - P(E \cup F) \geq P(E) + P(F) - 1$. ($n = 1$ is an equality.)

Step 2. (Induction.) Apply Step 1 to $E = E_1 \cdots E_n$, $F = E_{n+1}$:

$$\begin{aligned} P(E_1 \cdots E_{n+1}) &\geq P(E_1 \cdots E_n) + P(E_{n+1}) - 1 \\ &\geq \sum_{i=1}^{n} P(E_i) - (n-1) + P(E_{n+1}) - 1 = \sum_{i=1}^{n+1} P(E_i) - n. \end{aligned}$$

E.g. $P(E) = 0.9$, $P(F) = 0.8$ give $P(EF) \geq 0.7$ under any dependence. ■

Exercise 2.19 (Waiting for the r-th red ball)

An urn contains n red and m blue balls. They are withdrawn one at a time, at random, until a total of $r \le n$ red balls have been withdrawn. Find the probability that exactly k balls are withdrawn.

Solution.

Step 1. Take the $(n+m)!$ equally likely orders of all (labelled) balls. Exactly k are withdrawn iff the first $k-1$ positions hold exactly $r-1$ red and position k is red ($r \le k \le r+m$).

Step 2. Count: the balls in the first $k-1$ positions ($\binom{n}{r-1}\binom{m}{k-r}$) and their order ($(k-1)!$), the red ball at k ($n-r+1$), and the order of the rest ($(n+m-k)!$).

Step 3. Divide by $(n+m)!$, using $\binom{n+m}{k-1}^{-1} = (k-1)!\,(n+m-k+1)!/(n+m)!$:

$$P(k \text{ balls withdrawn}) = \frac{\binom{n}{r-1}\binom{m}{k-r}}{\binom{n+m}{k-1}} \cdot \frac{n-r+1}{n+m-k+1}, \qquad r \le k \le r+m.$$

(hypergeometric probability of $r-1$ red in $k-1$ draws, times that the next is red). ■

Exercise 2.20 (No uniform distribution on a countable set)

Let the sample space $S = \{s_1, s_2, \dots\}$ be countably infinite. Show that not all outcomes can be equally likely. Can all outcomes have positive probability?

Solution.

Step 1. By (ii) and (iii), $1 = P(S) = \sum_{i=1}^{\infty} P(\{s_i\})$.

Step 2. If all $P(\{s_i\}) = p$, the series is 0 ($p = 0$) or ∞ ($p > 0$), never 1.

Step 3. Yes: $P(E) = \sum_{i:s_i \in E} 2^{-i}$ is nonnegative, has $P(S) = 1$, and is countably additive (nonnegative series can be regrouped). ■

Lecture 3. Conditional Probability and Independence

3.1 Conditional probability

Motivation. Conditioning restricts the sample space to what is known, and the multiplication rule builds probabilities step by step.

Definition (Conditional probability)

If $P(F) > 0$, the conditional probability of E given F is $P(E \mid F) = P(EF)/P(F)$.

With equally likely outcomes $P(E \mid F) = |EF|/|F|$: shrink the sample space to F.

Proposition 3.1 (Multiplication rule)

$P(E_1E_2 \cdots E_n) = P(E_1)P(E_2 \mid E_1)P(E_3 \mid E_1E_2) \cdots P(E_n \mid E_1 \cdots E_{n-1})$.

Proof. Write each factor as a ratio; the product telescopes. ∎

Example 3.1

(a) 8 red and 4 white balls, two drawn without replacement: $P(\text{both red}) = \frac{8}{12} \cdot \frac{7}{11} = \frac{14}{33} = \binom{8}{2}/\binom{12}{2}$.
(b) A deck is split at random into four piles of 13; E_i: pile i has exactly one ace. Placing the aces one at a time (the second must fall in one of the 39 of the remaining 51 positions in other piles, and so on),

$$P(E_1E_2E_3E_4) = 1 \cdot \frac{39}{51} \cdot \frac{26}{50} \cdot \frac{13}{49} \approx 0.105,$$

Example 3.2 (Matching, recursively)

In Example 2.4, let P_N be the probability of no match and M the event that person 1 takes their own hat; $P_N = P(\text{no match} \mid M^c)\frac{N-1}{N}$. Given that person 1 took hat $j \neq 1$, either person j takes hat 1 (probability $\frac{1}{N-1}$) and the other $N-2$ have no match, or, treating hat 1 as j's own, the $N-1$ people have no match. So

$$P_N = \frac{N-1}{N}\Big(P_{N-1} + \frac{P_{N-2}}{N-1}\Big), \qquad \text{i.e.} \qquad P_N - P_{N-1} = -\frac{1}{N}(P_{N-1} - P_{N-2}).$$

With $P_1 = 0$, $P_2 = \frac{1}{2}$: $P_N - P_{N-1} = (-1)^N/N!$, hence $P_N = \sum_{r=0}^{N}(-1)^r/r!$.

3.2 Total probability and Bayes's formula

Motivation. Bayes's formula inverts $P(B \mid A_i)$ into $P(A_i \mid B)$, often counterintuitively.

Since $E = EF \cup EF^c$,

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)(1 - P(F)),$$

a weighted average. More generally:

Theorem 3.2 (Total probability and Bayes's formula)

If $F_1, \dots, F_n$ are mutually exclusive with union S, then

$$P(E) = \sum_{i=1}^{n} P(E \mid F_i)P(F_i), \qquad P(F_j \mid E) = \frac{P(E \mid F_j)P(F_j)}{\sum_i P(E \mid F_i)P(F_i)}.$$

$P(F_j)$ are **prior** and $P(F_j \mid E)$ **posterior** probabilities. In odds form,

$$\frac{P(H \mid E)}{P(H^c \mid E)} = \frac{P(H)}{P(H^c)} \cdot \frac{P(E \mid H)}{P(E \mid H^c)} :$$

evidence multiplies the odds by the **likelihood ratio**.

Example 3.3 (A screening test)

Sensitivity 0.95, false-positive rate 1%, prevalence 0.5%:

$$P(D \mid +) = \frac{0.95 \cdot 0.005}{0.95 \cdot 0.005 + 0.01 \cdot 0.995} \approx 0.323.$$

Only a third of positives are diseased: the many healthy produce more false positives than there are true ones.

Example 3.4 (Multiple choice)

A student knows the answer with probability p, otherwise guesses among m alternatives:

$$P(\text{knew} \mid \text{correct}) = \frac{p}{p + (1-p)/m} = \frac{mp}{1 + (m-1)p}.$$

With $m = 5$, $p = 1/2$ this is $5/6$.

Example 3.5 (Three cards)

Cards: red–red, black–black, red–black. A random card shows red on top. Not $1/2$: of three equally likely red faces, two belong to the red–red card, so $P(RR \mid R \text{ up}) = \frac{1/3}{1/3 \cdot 1 + 1/3 \cdot 1/2} = \frac{2}{3}$.

Example 3.6 (Searching for a plane)

A plane is equally likely in one of three regions; a search of region i misses it there with probability β_i. After an unsuccessful search of region 1,

$$P(R_1 \mid \text{miss}) = \frac{\beta_1/3}{\beta_1/3 + 1/3 + 1/3} = \frac{\beta_1}{\beta_1 + 2}, \qquad P(R_j \mid \text{miss}) = \frac{1}{\beta_1 + 2} \quad (j = 2, 3).$$

For fixed F, $Q(E) = P(E \mid F)$ is a probability, e.g. $P(E_1 \mid F) = P(E_1 \mid E_2F)P(E_2 \mid F) + P(E_1 \mid E_2^cF)P(E_2^c \mid F)$.

3.3 Independent events

Definition (Independence)

E and F are **independent** if $P(EF) = P(E)P(F)$. Events $E_1, \dots, E_n$ are (mutually) independent if $P(E_{i_1} \cdots E_{i_r}) = P(E_{i_1}) \cdots P(E_{i_r})$ for every subcollection.

Independence of E, F implies that of E, F^c: $P(EF^c) = P(E) - P(E)P(F) = P(E)P(F^c)$.

Example 3.7 (Pairwise does not imply mutual)

Two fair dice; E: sum 7, F: first die 4, G: second die 3. Each pair is independent, but $P(E \mid FG) = 1 \neq P(E)$. Also, sum 6 is *not* independent of F, while E is ($P(\text{sum} = 7 \mid \text{first} = k) = 1/6$).

Example 3.8 (Reliability)

n independent components, component i working with probability p_i: a parallel system works with probability $1 - \prod_i (1 - p_i)$, a series system $\prod_i p_i$.

Example 3.9 (E before F)

Independent trials, E and F disjoint on each. Conditioning on the first trial, $P(E \text{ first}) = P(E) + (1 - P(E) - P(F))P(E \text{ first})$, so

$$P(E \text{ occurs before } F) = \frac{P(E)}{P(E) + P(F)}.$$

Craps: a first roll of 7 or 11 wins, 2, 3, 12 loses; otherwise the sum k is the "point" and one rolls until k (win) or 7 (lose). With $p_k = P(\text{sum} = k)$,

$$P(\text{win}) = p_7 + p_{11} + \sum_{k \in \{4,5,6,8,9,10\}} p_k \frac{p_k}{p_k + p_7} = \frac{244}{495} \approx 0.493.$$

Example 3.10 (The problem of the points)

Probability of n successes before m failures in trials with success probability p: at least n successes in the first $m + n - 1$ trials, so

$$P_{n,m} = \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} p^k (1-p)^{m+n-1-k}.$$

(Pascal–Fermat: split the stakes of an interrupted game as $P_{n,m} : 1 - P_{n,m}$; e.g. 0.828 for $p = \frac{1}{2}$, $n = 4$, $m = 7$.)

Example 3.11 (Gambler's ruin)

A gambler with fortune i wins or loses one unit per play (probabilities p, $q = 1 - p$) until N or 0; P_i = probability of reaching N. Conditioning on the first play, $P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$,

$P_0 = 0$, so $P_{i+1} - P_i = (q/p)^i P_1$; summing and using $P_N = 1$,

$$P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq \frac{1}{2}, \\ \dfrac{i}{N}, & p = \frac{1}{2}, \end{cases}$$

As $N \to \infty$: ruin is certain if $p \leq \frac{1}{2}$; if $p > \frac{1}{2}$ survival has probability $1 - (q/p)^i$.

3.4 $P(\cdot \mid F)$ is a probability

So one may condition sequentially; E_1, E_2 are conditionally independent given F if $P(E_1E_2 \mid F) = P(E_1 \mid F)P(E_2 \mid F)$.

Example 3.12 (Laplace's rule of succession)

Coin $i \in \{0, \ldots, k\}$ shows heads with probability i/k; a random coin gives n heads in n flips. Flips are independent given the coin, so

$$P(\text{head on flip } n+1 \mid n \text{ heads}) = \frac{\sum_i (i/k)^{n+1}/(k+1)}{\sum_i (i/k)^n/(k+1)} \approx \frac{\int_0^1 x^{n+1}dx}{\int_0^1 x^n dx} = \frac{n+1}{n+2}$$

for large k. Unconditionally the flips are dependent: early heads indicate a good coin.

Example 3.13 (Updating sequentially)

If evidence $E_1, \ldots, E_k$ is conditionally independent given each hypothesis H_i,

$$P(H_i \mid E_1 \cdots E_k) \propto P(H_i) \prod_{j=1}^{k} P(E_j \mid H_i),$$

so each posterior is the next prior, in any order.

3.5 Selected exercises

Motivation. Ross's theoretical exercises (book numbering), solved by conditioning on a first or last event.

Exercise 3.7(a) (The colour that survives)

An urn contains $n \geq 1$ white and $m \geq 1$ black balls. The balls are withdrawn one at a time, at random, until only balls of one colour are left. Show that they are all white with probability $n/(n+m)$.

Solution.

Step 1. Continue the withdrawals until the urn is empty; the $(n+m)!$ orders of labelled balls are equally likely, and the experiment is an initial part of the order.

Step 2. The balls left form a nonempty final segment (both colours are present at the start), containing the last ball, and all have its colour. So they are white iff the last ball is white.

Step 3. Each ball is last in $(n+m-1)!$ orders, i.e. with probability $1/(n+m)$; so $P = n/(n+m)$. ■

Exercise 3.16 (An even number of successes)

Let P_n be the probability that n Bernoulli trials with success probability p result in an even number of successes (0 counts as even). Show that

$$P_n = p(1-P_{n-1}) + (1-p)P_{n-1} \quad (n \geq 1),$$

and deduce $P_n = (1+(1-2p)^n)/2$.

Solution.

Step 1. With S the successes in the first $n-1$ trials, the total is even iff trial n succeeds and S is odd, or fails and S is even; by independence

$$P_n = p\,P(S \text{ odd}) + (1-p)P(S \text{ even}) = p(1-P_{n-1}) + (1-p)P_{n-1}.$$

Step 2. I.e. $P_n = p + (1-2p)P_{n-1}$, with $P_0 = 1$. Inductively,

$$P_n = p + \frac{1-2p}{2} + \frac{(1-2p)^n}{2} = \frac{1+(1-2p)^n}{2},$$

■

Exercise 3.18 (No run of three heads)

Let Q_n be the probability that no run of three consecutive heads appears in n tosses of a fair coin. Show that

$$Q_n = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3} \quad (n \geq 3), \qquad Q_0 = Q_1 = Q_2 = 1,$$

and find Q_8.

Solution.

Step 1. Let R: no run of three heads; F_j: first tail at toss j ($j = 1, 2, 3$); H: tosses $1, 2, 3$ all heads. These partition the space, $P(F_j) = 2^{-j}$.

Step 2. $RH = \emptyset$. On F_j the first j tosses contain no run and toss j is a tail, so $RF_j = F_j \cap R_j$, R_j: no run among tosses $j+1, \ldots, n$.

Step 3. R_j is independent of F_j and $P(R_j) = Q_{n-j}$:

$$Q_n = \sum_{j=1}^{3} P(F_j)P(R_j) = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3}.$$

Step 4. $Q_3 = \frac{7}{8}$, $Q_4 = \frac{13}{16}$, $Q_5 = \frac{24}{32}$, $Q_6 = \frac{44}{64}$, $Q_7 = \frac{81}{128}$ and

$$Q_8 = \tfrac{1}{2}\cdot\tfrac{81}{128} + \tfrac{1}{4}\cdot\tfrac{44}{64} + \tfrac{1}{8}\cdot\tfrac{24}{32} = \frac{81+44+24}{256} = \frac{149}{256} \approx 0.582.$$

■

Exercise 3.21 (The ballot problem)

In an election, candidate A receives n votes and candidate B receives m votes, $n > m \geq 0$, and all $\binom{n+m}{n}$ orderings of the votes are equally likely. Let $P_{n,m}$ be the probability that A is strictly ahead throughout the count.
(a) Derive a recursion for $P_{n,m}$ in terms of $P_{n-1,m}$ and $P_{n,m-1}$ by conditioning on who receives the last vote.
(b) Prove that $P_{n,m} = (n-m)/(n+m)$.

Solution.

(a)

Step 1. Orderings ending with A correspond to orderings of $n-1$ A's and m B's, so the last vote is A's with probability $\binom{n+m-1}{n-1}/\binom{n+m}{n} = n/(n+m)$, and then the first $n+m-1$ votes are a uniform ordering of $n-1$ A's and m B's; likewise for B.

Step 2. Since $n > m$, A is ahead at the end, so A leads throughout iff A leads throughout the first $n+m-1$ votes. For $m \geq 1$

$$P_{n,m} = \frac{n}{n+m} P_{n-1,m} + \frac{m}{n+m} P_{n,m-1},$$

with $P_{k,k} = 0$ ($k \geq 1$; a tie at the end) and $P_{n,0} = 1$.

(b)

Step 1. Induction on $n+m$. $m = 0$: $P_{n,0} = 1 = n/n$; the formula gives 0 when $n = m$, matching $P_{k,k} = 0$.

Step 2. For $n > m \geq 1$, by (a) and the hypothesis (covering $P_{n-1,m}$ also when $n - 1 = m$),

$$\begin{aligned} P_{n,m} &= \frac{n}{n+m} \cdot \frac{n-1-m}{n+m-1} + \frac{m}{n+m} \cdot \frac{n-m+1}{n+m-1} \\ &= \frac{(n-m)(n+m) - (n-m)}{(n+m)(n+m-1)} = \frac{n-m}{n+m}, \end{aligned}$$

using $n(n-m-1) + m(n-m+1) = n^2 - m^2 - (n-m)$. ■

Lecture 4. Discrete Random Variables

4.1 Random variables and distribution functions

A **random variable** is a real function on S, with **distribution function** $F(x) = P(X \le x)$. It is **discrete** if it takes countably many values $x_1, x_2, \ldots$, with **probability mass function** $p(x) = P(X = x)$, $\sum_i p(x_i) = 1$ and step function $F(a) = \sum_{x_i \le a} p(x_i)$.

Proposition 4.1 (Properties of F)

F is nondecreasing, right-continuous, $F(-\infty) = 0$ and $F(\infty) = 1$. Moreover $P(a < X \le b) = F(b) - F(a)$ and $P(X < b) = \lim_{x \uparrow b} F(x)$.

Proof. Monotonicity: Proposition 2.1(b). As $x_n \downarrow b$, $\{X \le x_n\} \downarrow \{X \le b\}$; as $x_n \uparrow \infty$, $\uparrow S$; as $x_n \uparrow b$, $\uparrow \{X < b\}$. Apply Proposition 2.3. ■

Example 4.1 (Coupon collecting)

Each purchase gives one of N equally likely coupon types, independently; T is the number of purchases to collect all. With A_j: type j missing after n purchases,

$$P(T > n) = P\Big(\bigcup_j A_j\Big) = \sum_{i=1}^{N-1} (-1)^{i+1} \binom{N}{i} \Big(\frac{N-i}{N}\Big)^n,$$

and $P(T = n) = P(T > n-1) - P(T > n)$.

4.2 Expectation

Motivation. The expectation is the long-run average, a one-number summary underlying fair prices, variance and the laws of large numbers.

Definition (Expected value)

$E[X] = \sum_x x p(x)$, provided the sum converges absolutely.

It is the centre of gravity of the pmf; $E[I_A] = P(A)$ for an indicator.

Proposition 4.2

(a) $E[g(X)] = \sum_x g(x)p(x)$.
(b) $E[aX + b] = aE[X] + b$.
(c) On a countable sample space, $E[X] = \sum_{s \in S} X(s)P(\{s\})$, and hence $E[\sum_i X_i] = \sum_i E[X_i]$.

Proof. (a) Group the values x by $y = g(x)$: $\sum_y y P(g(X) = y) = \sum_y \sum_{x: g(x)=y} g(x)p(x) = \sum_x g(x)p(x)$.

(b) is a special case.

(c) Group outcomes s by $X(s)$; linearity of the sum over s gives the second claim. ■

In general $E[g(X)] \ne g(E[X])$.

Example 4.2 (Which question first?)

A contestant answers question 1 or 2 first and continues only if correct; question i is correct with probability p_i independently and worth V_i. Order 1–2 gives $p_1(1-p_2)V_1+p_1p_2(V_1+V_2) = p_1V_1 + p_1p_2V_2$, order 2–1 gives $p_2V_2 + p_1p_2V_1$; so 1 first iff

$$\frac{p_1V_1}{1-p_1} > \frac{p_2V_2}{1-p_2}.$$

Example 4.3 (The friendship paradox)

Pick a random friendship and one of its ends; that person has D^* friends, with $P(D^* = k) = k\,P(D = k)/E[D]$, D the degree of a uniform person (a person with k friends is k times as likely reached). Hence

$$E[D^*] = \frac{E[D^2]}{E[D]} \geq E[D],$$

with equality only for equal degrees: on average your friends have more friends than you.

4.3 Variance

Definition (Variance)

With $\mu = E[X]$, $\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$. The standard deviation is $\sqrt{\mathrm{Var}(X)}$.

(Expand the square.) So $E[X^2] \geq (E[X])^2$, and $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$. Fair die: $E[X] = 7/2$, $E[X^2] = 91/6$, $\mathrm{Var}(X) = 35/12$.

4.4 Bernoulli and binomial random variables

$X \sim \mathrm{Bin}(n,p)$ counts successes in n independent trials with success probability p:

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

(each of the $\binom{n}{k}$ sequences with k successes has probability $p^k(1-p)^{n-k}$). $\mathrm{Bin}(1,p)$ is Bernoulli.

Proposition 4.3

If $X \sim \mathrm{Bin}(n,p)$ then $E[X^k] = np\,E[(Y+1)^{k-1}]$ with $Y \sim \mathrm{Bin}(n-1,p)$. Hence $E[X] = np$ and $\mathrm{Var}(X) = np(1-p)$.

Proof. Using $i\binom{n}{i} = n\binom{n-1}{i-1}$,

$$E[X^k] = \sum_{i=1}^{n} i^{k-1}\, n \binom{n-1}{i-1} p^i (1-p)^{n-i} = np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j}.$$

$k = 1$ gives np; $k = 2$ gives $E[X^2] = np\{(n-1)p+1\}$, so $\mathrm{Var}(X) = np(1-p)$. ∎

Since $p(k)/p(k-1) = \frac{(n-k+1)p}{k(1-p)}$, the pmf increases while $k < (n+1)p$, then decreases: mode $\lfloor (n+1)p \rfloor$. The ratio also gives a stable cdf recursion.

Example 4.4 (Chuck-a-luck)

Bet 1 on a number; three dice are rolled; win i if it appears $i \geq 1$ times, else lose 1. Appearances are Bin$(3, 1/6)$, so the expected gain is

$$-\tfrac{125}{216} + 1 \cdot \tfrac{75}{216} + 2 \cdot \tfrac{15}{216} + 3 \cdot \tfrac{1}{216} = -\tfrac{17}{216} \approx -0.079.$$

Example 4.5 (k-out-of-n systems)

n components work independently with probability p; the system works if at least half do. $P_5 = \binom{5}{3}p^3q^2 + 5p^4q + p^5$, $P_3 = 3p^2q + p^3$, and $P_5 - P_3 = 3p^2(p-1)^2(2p-1)$: five components are better iff $p > \frac{1}{2}$.

4.5 The Poisson random variable

Motivation. Rare-event counts are approximately Poisson, the limit of binomials with large n and small p.

$X \sim \text{Poisson}(\lambda)$ if $p(k) = e^{-\lambda}\lambda^k/k!$, $k \geq 0$; $E[X] = \text{Var}(X) = \lambda$ (shift trick: $E[X] = \lambda \sum_j e^{-\lambda}\lambda^j/j!$, $E[X^2] = \lambda E[X+1]$).

Proposition 4.4 (Poisson approximation)

If $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed, then $\binom{n}{k}p^k(1-p)^{n-k} \to e^{-\lambda}\lambda^k/k!$.

Proof. $\binom{n}{k}p^k(1-p)^{n-k} = \dfrac{n(n-1)\cdots(n-k+1)}{n^k} \cdot \dfrac{\lambda^k}{k!} \cdot \dfrac{(1-\lambda/n)^n}{(1-\lambda/n)^k}$, and the three factors tend to 1, $\lambda^k/k!$ and $e^{-\lambda}$. ■

So counts of rare events among many trials are approximately Poisson, also for *weakly dependent* trials with small unequal probabilities, $\lambda = \sum p_i$ (the Poisson paradigm).

Example 4.6 (Poisson paradigm)

(a) Matching: the weakly dependent events "person i gets own hat" have probability $1/N$, so matches are approximately Poisson(1) (cf. Example 2.4).
(b) Birthdays: the $\binom{n}{2}$ weakly dependent pair events have probability $1/365$, so $P(\text{no shared birthday}) \approx \exp\{-n(n-1)/730\}$; $n = 23$ gives 0.500 (exact 0.493).

Example 4.7 (Poisson process)

Events occur in time with (i) independent counts on disjoint intervals, (ii) one event in an interval of length h with probability $\lambda h + o(h)$, two or more with $o(h)$. Splitting $[0, t]$ into n pieces, $N(t) \approx \text{Bin}(n, \lambda t/n)$, and as $n \to \infty$

$$P(N(t) = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}.$$

4.6 Other discrete distributions

Geometric. Trials up to the first success: $p(k) = (1-p)^{k-1}p$, $k \geq 1$, $P(X > k) = (1-p)^k$. With $q = 1-p$, conditioning on the first trial, $E[X] = p + q(1 + E[X])$ gives $E[X] = 1/p$, and $E[X^2] = p + q\,E[(1+X)^2]$ gives $E[X^2] = (2-p)/p^2$, $\mathrm{Var}(X) = q/p^2$.

Negative binomial. Trials needed for r successes ($r-1$ successes in the first $k-1$ trials, then a success):

$$p(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \geq r, \qquad E[X] = \frac{r}{p}, \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

(A sum of r independent geometrics; see Lecture 7.)

Example 4.8 (Banach match problem)

A smoker has two boxes of N matches and picks one at random each time. When a box is first found empty, what is the probability the other has k matches? With "left box" a success, the left box is found empty with k on the right when success $N+1$ occurs at trial $2N-k+1$; doubling for symmetry, $2\binom{2N-k}{N}(\frac{1}{2})^{2N-k+1} = \binom{2N-k}{N}(\frac{1}{2})^{2N-k}$.

Hypergeometric. White balls X among n drawn without replacement from N, m white:

$$P(X=i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}, \qquad E[X] = \frac{nm}{N}, \qquad \mathrm{Var}(X) = n\frac{m}{N}\Big(1-\frac{m}{N}\Big)\frac{N-n}{N-1}.$$

With $p = m/N$: the binomial mean, and variance with finite-population correction $(N-n)/(N-1)$. For $N \gg n$, $X \approx \mathrm{Bin}(n, m/N)$.

Example 4.9 (Capture–recapture)

Tag m animals, later catch n, of which i are tagged. $P_i(N) = \binom{m}{i}\binom{N-m}{n-i}/\binom{N}{n}$ satisfies

$$\frac{P_i(N)}{P_i(N-1)} = \frac{(N-m)(N-n)}{N(N-m-n+i)} \geq 1 \iff N \leq \frac{mn}{i},$$

so the maximum likelihood estimate is $\lfloor mn/i \rfloor$ (setting $i/n = m/N$).

Zeta (Zipf). $p(k) = C/k^{\alpha+1}$, $k \geq 1$, $C = 1/\zeta(\alpha+1)$; heavy-tailed frequencies such as word counts.

4.7 Expected value of sums

By Proposition 4.2(c), $E[X_1 + \cdots + X_n] = \sum E[X_i]$ for *any* random variables. Writing counts as sums of indicators is the main trick.

Example 4.10

(a) The sum of n fair dice has mean $7n/2$.
(b) If trial i succeeds with probability p_i, the expected number of successes is $\sum p_i$.

(c) Hypergeometric: with I_j indicating that white ball j is drawn, $E[X] = m \cdot n/N$ (Example 2.1(b)).

Example 4.11 (Variance of the number of successes)

$X = \sum_i I_i$ gives $E[X^2] = \sum_i p_i + \sum_{i \neq j} P(I_i = I_j = 1)$. Independent trials: $E[X^2] = np + n(n-1)p^2$, $\mathrm{Var}(X) = np(1-p)$. Hypergeometric: $P(I_i = I_j = 1) = \frac{m(m-1)}{N(N-1)}$, giving the variance above.

4.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): re-indexing binomial sums, conditioning on a count, and links among Poisson, gamma, binomial and negative binomial.

Exercise 4.10 (A reciprocal moment of the binomial)

Let $X \sim \mathrm{Bin}(n, p)$ with $0 < p \leq 1$. Show that

$$E\Big[\frac{1}{X+1}\Big] = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution.

Step 1. For $0 \leq i \leq n$,

$$\frac{1}{i+1}\binom{n}{i} = \frac{n!}{(i+1)!\,(n-i)!} = \frac{1}{n+1}\binom{n+1}{i+1}.$$

Step 2. With $q = 1-p$ and Proposition 4.2(a),

$$\begin{aligned} E\Big[\frac{1}{X+1}\Big] &= \sum_{i=0}^{n} \frac{1}{i+1}\binom{n}{i} p^i q^{n-i} = \frac{1}{(n+1)p} \sum_{i=0}^{n} \binom{n+1}{i+1} p^{i+1} q^{n-i} \\ &= \frac{1}{(n+1)p} \sum_{j=1}^{n+1} \binom{n+1}{j} p^j q^{n+1-j} = \frac{(p+q)^{n+1} - q^{n+1}}{(n+1)p}, \end{aligned}$$

by the binomial theorem (Theorem 1.3) without the $j = 0$ term; $p + q = 1$. ∎

Exercise 4.25 (Thinning a Poisson count)

The number N of events in a fixed period is Poisson(λ), and each event is counted with probability p, independently of the other events and of N. Show that the number M of counted events is Poisson(λp). If $\lambda = 10$ uranium deposits are expected in an area and each is discovered with probability $1/50$, find the probability that at most one deposit is discovered.

Solution.

Step 1. Given $N = n$, $M \sim \mathrm{Bin}(n, p)$: $P(M = k \mid N = n) = \binom{n}{k} p^k q^{n-k}$ $(k \leq n)$, $q = 1-p$.

Step 2. By total probability over the countable partition $\{N = n\}$ (Theorem 3.2),

$$P(M = k) = \sum_{n=k}^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \cdot \frac{n!}{k!\,(n-k)!} p^k q^{n-k}$$

$$= e^{-\lambda}\frac{(\lambda p)^k}{k!}\sum_{j=0}^{\infty}\frac{(\lambda q)^j}{j!} = e^{-\lambda}\frac{(\lambda p)^k}{k!}e^{\lambda q} = e^{-\lambda p}\frac{(\lambda p)^k}{k!}.$$

Step 3. (Intuition: $N \approx \text{Bin}(n, \lambda/n)$, so $M \approx \text{Bin}(n, \lambda p/n) \to \text{Poisson}(\lambda p)$ by Proposition 4.4.)

Step 4. Here $\lambda p = 0.2$, so $P(M \le 1) = e^{-0.2}(1 + 0.2) \approx 0.982$. ■

Exercise 4.26 (Poisson tails as gamma integrals)

Prove that for $\lambda > 0$ and $n = 0, 1, 2, \ldots,$

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx.$$

Solution.

Step 1. Let $I_n = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx$; $I_0 = e^{-\lambda}$.

Step 2. For $n \ge 1$ integrate by parts ($u = x^n/n!$, $dv = e^{-x}dx$; $x^n e^{-x} \to 0$):

$$I_n = \left[-e^{-x}\frac{x^n}{n!}\right]_{\lambda}^{\infty} + \int_{\lambda}^{\infty} e^{-x}\frac{x^{n-1}}{(n-1)!}\,dx = e^{-\lambda}\frac{\lambda^n}{n!} + I_{n-1}.$$

Step 3. Induct on n. Equivalently, $P(X \le n) = P(Y > \lambda)$ for $X \sim \text{Poisson}(\lambda)$, $Y \sim \text{Gamma}(n + 1, 1)$: in a rate-one Poisson process, at most n events by time λ iff event $n + 1$ comes after λ. ■

Exercise 4.28 (Negative binomial versus binomial)

Let X be negative binomial with parameters (r, p) and $Y \sim \text{Bin}(n, p)$. Show that $P(X > n) = P(Y < r)$.

Solution.

Step 1. In one sequence of independent trials, let X be the trial of the r-th success and Y the successes among the first n; these have the stated laws (Section 4.6), so show $\{X > n\} = \{Y < r\}$.

Step 2. $Y \ge r$ iff the first n trials contain r successes iff $X \le n$. Take complements. ■

Lecture 5. Continuous Random Variables

5.1 Densities

X is **continuous** if $P(X \in B) = \int_B f(x)\,dx$ for a nonnegative **density** f and all (reasonable) B. Then $\int f = 1$, $P(X = a) = 0$, $F(a) = \int_{-\infty}^{a} f$, $F' = f$ where f is continuous, and $P(a - \frac{\epsilon}{2} \le X \le a + \frac{\epsilon}{2}) \approx \epsilon f(a)$.

Example 5.1

(a) $f(x) = C(4x - 2x^2)$ on $(0, 2)$: $1 = 8C/3$, so $C = 3/8$, and $P(X > 1) = 1/2$ by symmetry.
(b) If $Y = 2X$ then $F_Y(a) = F_X(a/2)$ and $f_Y(a) = \frac{1}{2} f_X(a/2)$.

5.2 Expectation and variance

Motivation. Sums become integrals; the tail formula $E[X] = \int_0^\infty P(X > x)dx$ is often quickest.

$E[X] = \int x f(x)\,dx$; $\mathrm{Var}(X) = E[X^2] - \mu^2$.

Lemma 5.1 (Tail formula)

If $Y \ge 0$, then $E[Y] = \int_0^\infty P(Y > y)\,dy$.

Proof. $\int_0^\infty P(Y > y)\,dy = \int_0^\infty \int_y^\infty f_Y(x)\,dx\,dy = \int_0^\infty \left(\int_0^x dy\right) f_Y(x)\,dx = \int_0^\infty x f_Y(x)\,dx$. ■

(Discrete analogue: $E[N] = \sum_{k \ge 1} P(N \ge k)$, $N \ge 0$ integer.)

Proposition 5.2

$E[g(X)] = \int g(x) f(x)\,dx$.

Proof. For $g \ge 0$, by the lemma, $E[g(X)] = \int_0^\infty P(g(X) > y)\,dy = \int_0^\infty \int_{x:g(x)>y} f(x)\,dx\,dy = \int \left(\int_0^{g(x)} dy\right) f(x)\,dx = \int g(x) f(x)\,dx$. In general split $g = g^+ - g^-$. ■

Example 5.2 (Breaking a stick)

A unit stick is broken at a uniform U. The piece containing a fixed $p \in [0, 1]$ has length $L_p(U) = 1 - U$ $(U < p)$ or U $(U > p)$:

$$E[L_p(U)] = \int_0^p (1 - u)\,du + \int_p^1 u\,du = \frac{1}{2} + p(1 - p).$$

This is $\ge \frac{1}{2}$, maximal at $p = \frac{1}{2}$: longer pieces are more likely to contain p (cf. Example 4.3).

Example 5.3 (When to leave)

Travel time X has density f, cdf F; arriving s early costs cs, s late ks. Leaving t minutes

ahead costs on average

$$E[C_t(X)] = \int_0^t c(t-x)f(x)\,dx + \int_t^\infty k(x-t)f(x)\,dx, \qquad \frac{d}{dt}E[C_t(X)] = cF(t) - k\{1-F(t)\}.$$

The optimum is $F(t^*) = k/(k+c)$, the median for $k = c$; likewise $E|X-a|$ is minimized at the median.

5.3 The uniform distribution

$X \sim U(\alpha, \beta)$ has density $1/(\beta-\alpha)$ on (α, β). Then $E[X] = (\alpha+\beta)/2$ and $\mathrm{Var}(X) = (\beta-\alpha)^2/12$.

Example 5.4 (Bertrand's paradox)

Is a "random chord" of a circle of radius r longer than the side $\sqrt{3}r$ of the inscribed triangle? A chord at distance D from the centre has length $2\sqrt{r^2-D^2} > \sqrt{3}r$ iff $D < r/2$.
(i) $D \sim U(0, r)$ (perpendicular through a random point of a radius): $1/2$.
(ii) One endpoint fixed, angle to the tangent $\theta \sim U(0, \pi)$: length $2r\sin\theta > \sqrt{3}r$ iff $\theta \in (\pi/3, 2\pi/3)$: $1/3$. "Random chord" is undefined until the experiment is specified.

5.4 Normal random variables

$X \sim N(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \qquad -\infty < x < \infty.$$

It integrates to 1 since $I^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi$, $I = \int e^{-x^2/2}\,dx$. Since $aX + b \sim N(a\mu+b, a^2\sigma^2)$, $Z = (X-\mu)/\sigma$ is **standard normal** (cdf Φ, $\Phi(-x) = 1 - \Phi(x)$), with $E[Z] = 0$ and $E[Z^2] = 1$ (by parts); so $E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, and

$$P(X \le a) = \Phi\Big(\frac{a-\mu}{\sigma}\Big).$$

$P(|Z| < 1, 2, 3) \approx 0.683, 0.954, 0.997$; $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$; $1 - \Phi(x) \approx e^{-x^2/2}/(x\sqrt{2\pi})$ for large x.

Example 5.5 (Detecting a signal)

A bit is sent as ± 2, received as $R = \pm 2 + N$, $N \sim N(0,1)$, decoded as 1 iff $R \ge 0.5$. Error probabilities: $P(N < -1.5) \approx 0.067$ and $P(N \ge 2.5) \approx 0.006$.

Theorem 5.3 (De Moivre–Laplace)

If $S_n \sim \mathrm{Bin}(n, p)$, then for $a < b$, $P\Big(a \le \dfrac{S_n - np}{\sqrt{np(1-p)}} \le b\Big) \to \Phi(b) - \Phi(a)$ as $n \to \infty$.

A special case of the CLT (Lecture 8), good when $np(1-p) \ge 10$. Use the **continuity correction** $P(S_n = i) = P(i - \frac{1}{2} < S_n < i + \frac{1}{2})$; for small p use Poisson.

Example 5.6

(a) 40 fair flips: $P(S=20) \approx \Phi(0.5/\sqrt{10}) - \Phi(-0.5/\sqrt{10}) \approx 0.1256$ (exact 0.1254).
(b) 450 admitted, each enrolling with probability 0.3, room for 150: $S \sim \mathrm{Bin}(450, 0.3)$, mean 135, variance 94.5, $P(S \geq 151) = P(S \geq 150.5) \approx 1 - \Phi(15.5/\sqrt{94.5}) = 1 - \Phi(1.59) \approx 0.055$ (exact: 0.057).

5.5 Exponential random variables

$X \sim \mathrm{Exp}(\lambda)$: $f(x) = \lambda e^{-\lambda x}$, $P(X > x) = e^{-\lambda x}$, $x \geq 0$; $E[X] = 1/\lambda$, $\mathrm{Var}(X) = 1/\lambda^2$. In a Poisson process (Example 4.7) the first event comes after t iff $N(t) = 0$, probability $e^{-\lambda t}$.

Proposition 5.4 (Memorylessness)

$P(X > s+t \mid X > t) = P(X > s)$ for all $s, t \geq 0$ holds for $X \sim \mathrm{Exp}(\lambda)$, and the exponential distributions are the only continuous distributions on $[0, \infty)$ with this property.

Proof. $e^{-\lambda(s+t)}/e^{-\lambda t} = e^{-\lambda s}$. Conversely $\bar{F}(x) = P(X > x)$ satisfies $\bar{F}(s+t) = \bar{F}(s)\bar{F}(t)$, so $\bar{F}(m/n) = \bar{F}(1/n)^m$, $\bar{F}(1) = \bar{F}(1/n)^n$, hence $\bar{F}(x) = \bar{F}(1)^x$ for rational x and, by right continuity, all $x \geq 0$: $\bar{F}(x) = e^{-\lambda x}$, $\lambda = -\log \bar{F}(1)$. ∎

Example 5.7 (Two clerks)

A customer arrives while Jones and Brown are being served by two clerks and is served when either finishes; service times are i.i.d. exponential. When the first leaves, the other's remaining time is again exponential (memorylessness), so the newcomer is last with probability 1/2.

Hazard rate. For a lifetime X, $\lambda(t) = f(t)/\bar{F}(t)$ ($\lambda(t)\,dt$: failure probability in dt at age t). Since $\lambda(t) = -\frac{d}{dt}\log \bar{F}(t)$,

$$\bar{F}(t) = \exp\Big\{-\int_0^t \lambda(s)\,ds\Big\},$$

so λ determines F; constant hazard is exponential, increasing hazard models ageing.

Example 5.8

If a smoker's death rate is twice a non-smoker's at every age, the smoker's survival from A to B is $\exp\{-2\int_A^B \lambda_N\}$, the *square* of the non-smoker's: 0.76 (age 50 to 60) becomes 0.578.

5.6 Other continuous distributions

Gamma. $f(x) = \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}/\Gamma(\alpha)$, $x > 0$, with $\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\,dy = (\alpha-1)\Gamma(\alpha-1)$, $\Gamma(n) = (n-1)!$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$; $E[X] = \alpha/\lambda$, $\mathrm{Var}(X) = \alpha/\lambda^2$. The n-th event time T_n of a Poisson process is Gamma(n, λ):

$$P(T_n \leq t) = P(N(t) \geq n) = \sum_{j \geq n} e^{-\lambda t}\frac{(\lambda t)^j}{j!},$$

whose derivative in t telescopes to the density $\lambda e^{-\lambda t}(\lambda t)^{n-1}/(n-1)!$. **Chi-squared** with n d.f. is Gamma$(n/2, 1/2)$.

Weibull. $F(x) = 1 - \exp\{-((x-\nu)/\alpha)^\beta\}$, $x > \nu$; weakest-link lifetimes, hazard $\propto (x-\nu)^{\beta-1}$.

Cauchy. $f(x) = \frac{1}{\pi}\frac{1}{1+(x-\theta)^2}$. A flashlight at distance 1 from a wall at uniform angle $\theta \in (-\pi/2, \pi/2)$ lights $X = \tan\theta$, with $P(X \le x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$. No mean: $\int |x|/(1+x^2)\,dx = \infty$.

Beta. $f(x) = x^{a-1}(1-x)^{b-1}/B(a,b)$ on $(0,1)$, $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $E[X] = a/(a+b)$, $\mathrm{Var}(X) = ab/\{(a+b)^2(a+b+1)\}$; random proportions ($a = b = 1$: uniform).

Pareto. $Y = ae^X$, $X \sim \mathrm{Exp}(\lambda)$, has $P(Y > y) = (a/y)^\lambda$, $y \ge a$; given $Y > y_0$ it is Pareto with $a \to y_0$.

5.7 The distribution of a function of a random variable

Motivation. The density of $Y = g(X)$: differentiate the cdf; for monotone g this gives a Jacobian formula.

Theorem 5.5

If g is strictly monotone and differentiable on the range of X, then $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right|$$

for y in the range of g, and 0 otherwise.

Proof. Increasing g: $F_Y(y) = F_X(g^{-1}(y))$; differentiate. Decreasing: $F_Y(y) = 1 - F_X(g^{-1}(y))$ and $(g^{-1})' < 0$. ■

Example 5.9

(a) $Y = X^2$: $F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$, $f_Y(y) = \frac{1}{2\sqrt{y}}\{f_X(\sqrt{y}) + f_X(-\sqrt{y})\}$; for $X = Z$, $\frac{1}{\sqrt{2\pi y}}e^{-y/2}$, the χ^2_1 density (so $\Gamma(\frac{1}{2}) = \sqrt{\pi}$).
(b) $|X|$ has density $f_X(y) + f_X(-y)$ for $y > 0$.
(c) Lognormal: $Y = e^X$, $X \sim N(\mu, \sigma^2)$, $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma y}\exp\{-(\log y - \mu)^2/(2\sigma^2)\}$, $y > 0$ (multiplicative price changes).
(d) For continuous strictly increasing F, $F(X) \sim U(0,1)$ and $F^{-1}(U) \sim F$: the basis of simulation.

5.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): tail moments, a variance bound, normal moments, and the probability integral transform for general F.

Exercise 5.5 (Moments from the tail)

Use Lemma 5.1 to show that for a nonnegative random variable X and $n \ge 1$,

$$E[X^n] = \int_0^\infty nx^{n-1}P(X > x)\,dx.$$

Deduce $E[X^2]$ for $X \sim \text{Exp}(\lambda)$.

Solution.

Step 1. By Lemma 5.1, $E[X^n] = \int_0^\infty P(X^n > t)\,dt$.

Step 2. Substitute $t = x^n$ ($dt = nx^{n-1}dx$); for $X, x \geq 0$, $X^n > x^n$ iff $X > x$.

Step 3. For $\text{Exp}(\lambda)$,

$$E[X^2] = \int_0^\infty 2xe^{-\lambda x}\,dx = \frac{2}{\lambda^2},$$

so $\text{Var}(X) = 1/\lambda^2$ (Section 5.5). ■

Exercise 5.8 (Variance of a bounded variable)

Let $P(0 \leq X \leq c) = 1$ for some $c > 0$. Show that $\text{Var}(X) \leq c^2/4$.

Solution.

Step 1. $X^2 \leq cX$ on $\{0 \leq X \leq c\}$, so $E[X^2] \leq cE[X]$.

Step 2. With $\alpha = E[X]/c \in [0,1]$,

$$\text{Var}(X) = E[X^2] - (E[X])^2 \leq cE[X] - (E[X])^2 = c^2\alpha(1-\alpha) \leq \frac{c^2}{4},$$

as $\alpha(1-\alpha) = \frac{1}{4} - (\alpha - \frac{1}{2})^2$.

Step 3. Equality for $P(X = 0) = P(X = c) = \frac{1}{2}$. ■

Exercise 5.11 (Stein's identity and normal moments)

Let $Z \sim N(0,1)$ with density ϕ, and let g be differentiable with $E|g'(Z)| < \infty$, $E|Zg(Z)| < \infty$ and $g(x)\phi(x) \to 0$ as $|x| \to \infty$.
(a) Show that $E[g'(Z)] = E[Zg(Z)]$.
(b) Show that $E[Z^{n+1}] = nE[Z^{n-1}]$ for $n \geq 1$.
(c) Find $E[Z^4]$.

Solution.

(a)

Step 1. $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ satisfies $\phi'(x) = -x\phi(x)$.

Step 2. For $A, B > 0$, by parts,

$$\int_{-A}^{B} g'(x)\phi(x)\,dx = g(B)\phi(B) - g(-A)\phi(-A) + \int_{-A}^{B} xg(x)\phi(x)\,dx.$$

Step 3. As $A, B \to \infty$ the boundary terms vanish and the integrals converge ($g'\phi$, $xg\phi$ integrable); use Proposition 5.2.

(b) $g(x) = x^n$: all moments are finite ($|x|^k\phi(x) \leq C_k e^{-x^2/4}$) and $x^n\phi(x) \to 0$, so $E[Z^{n+1}] = nE[Z^{n-1}]$.

(c) $E[Z^2] = 1$, $E[Z^4] = 3E[Z^2] = 3$; generally $E[Z^{2k}] = (2k-1)(2k-3)\cdots 1$, odd moments 0. ■

Exercise 5.29 (The probability integral transform)

Let X be a continuous random variable with distribution function F. Show that $Y = F(X)$ is uniformly distributed on $(0, 1)$. (Example 5.9(d) assumed F strictly increasing; here F may be constant on intervals.)

Solution.

Step 1. F is continuous, nondecreasing, $F(-\infty) = 0$, $F(\infty) = 1$ (Proposition 4.1). Show $P(Y \leq y) = y$ for $0 < y < 1$ and $P(Y \leq 0) = 0$.

Step 2. For $y \in (0, 1)$, $L = \{x : F(x) = y\}$ is nonempty (IVT), closed, bounded above ($F \to 1 > y$); let $q = \max L$, $F(q) = y$.

Step 3. $\{F(X) \leq y\} = \{X \leq q\}$: $X \leq q$ gives $F(X) \leq y$; $X > q$ gives $F(X) \geq y$ and $F(X) \neq y$ ($X \notin L$), so $F(X) > y$.

Step 4. Hence $P(Y \leq y) = P(X \leq q) = F(q) = y$.

Step 5. If $F > 0$ everywhere, $P(Y \leq 0) = 0$. Otherwise let $a = \max\{x : F(x) = 0\}$ (nonempty, closed, bounded above); as in Step 3, $\{F(X) = 0\} = \{X \leq a\}$, so $P(Y \leq 0) = F(a) = 0$. ■

Lecture 6. Jointly Distributed Random Variables

6.1 Joint distribution functions

The **joint cdf** of X, Y is $F(a,b) = P(X \le a, Y \le b)$; the marginals are $F_X(a) = F(a,\infty)$ and $F_Y(b) = F(\infty, b)$, and

$$P(a_1 < X \le a_2,\ b_1 < Y \le b_2) = F(a_2,b_2) - F(a_1,b_2) - F(a_2,b_1) + F(a_1,b_1).$$

Discrete: joint pmf $p(x,y) = P(X=x, Y=y)$, $p_X(x) = \sum_y p(x,y)$. **Jointly continuous** with density f: $P((X,Y)\in C) = \iint_C f$, $f = \partial^2 F/\partial a\,\partial b$, $f_X(x) = \int f(x,y)\,dy$; likewise for n variables.

Example 6.1

$f(x,y) = 2e^{-x}e^{-2y}$, $x,y>0$: $P(X>1, Y<1) = e^{-1}(1-e^{-2}) \approx 0.318$ and

$$P(X<Y) = \int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,dx\,dy = \int_0^\infty 2e^{-2y}(1-e^{-y})\,dy = 1 - \tfrac{2}{3} = \tfrac{1}{3}.$$

Example 6.2 (A random point in a disc)

Uniform on the disc of radius R: $f_X(x) = \frac{2\sqrt{R^2-x^2}}{\pi R^2}$, $|x| \le R$ (not uniform); the distance D to the centre has $P(D \le a) = a^2/R^2$, $f_D(a) = 2a/R^2$, $E[D] = 2R/3$.

Example 6.3 (Multinomial distribution)

n independent trials with outcome probabilities $p_1, \dots, p_r$; the counts satisfy

$$P(X_1 = n_1, \dots, X_r = n_r) = \frac{n!}{n_1! \cdots n_r!} p_1^{n_1} \cdots p_r^{n_r}, \qquad \sum n_i = n,$$

($\binom{n}{n_1,\dots,n_r}$ sequences of probability $\prod p_i^{n_i}$). Marginally $X_i \sim \mathrm{Bin}(n, p_i)$.

6.2 Independent random variables

Motivation. Independence factors the joint law into marginals.

X and Y are **independent** if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all A, B; equivalently $F(a,b) = F_X(a)F_Y(b)$, or $p(x,y) = p_X(x)p_Y(y)$, or $f(x,y) = f_X(x)f_Y(y)$.

Proposition 6.1 (Factorization)

Jointly continuous (or discrete) X, Y are independent iff the joint density (pmf) factors as $f(x,y) = h(x)g(y)$ for all x, y.

Proof. If $f = hg$, with $C_1 = \int h$, $C_2 = \int g$: $C_1C_2 = 1$, $f_X = hC_2$, $f_Y = gC_1$, so $f_X f_Y = f$. ∎

It must hold on the whole plane: $f = 24xy$ on $\{x, y > 0, x+y < 1\}$ is not a product (non-rectangular support).

Example 6.4 (Thinning a Poisson count)

$N \sim \text{Poisson}(\lambda)$ customers, each independently a man with probability p; X, Y the numbers of men and women:

$$P(X=i, Y=j) = e^{-\lambda}\frac{\lambda^{i+j}}{(i+j)!}\binom{i+j}{i}p^i q^j = e^{-\lambda p}\frac{(\lambda p)^i}{i!}\cdot e^{-\lambda q}\frac{(\lambda q)^j}{j!}:$$

independent Poissons with means λp, λq (for fixed n they would be strongly dependent).

Example 6.5 (Meeting)

Two people arrive independently uniformly in a 30-minute window and wait 10 minutes: they meet iff $|X-Y| \le 10$, $P = 1-(20/30)^2 = 5/9$.

Example 6.6 (Buffon's needle)

A needle of length L falls on lines $D \ge L$ apart; the centre's distance to the nearest line is $X \sim U(0, D/2)$ and the acute angle $\theta \sim U(0, \pi/2)$, independent. It crosses iff $X < \frac{L}{2}\cos\theta$:

$$P(\text{cross}) = \frac{4}{\pi D}\int_0^{\pi/2}\int_0^{(L/2)\cos\theta} dx\, d\theta = \frac{2L}{\pi D}.$$

(a Monte Carlo estimate of π).

Example 6.7 (Characterizing the normal)

If X, Y are i.i.d. with differentiable density f and $f(x)f(y) = g(x^2+y^2)$, then $\frac{f'(x)}{2xf(x)} = \frac{g'(x^2+y^2)}{g(x^2+y^2)}$ for all x, y; the left side is free of y, hence both are constant: $f'(x)/f(x) = cx$, $f(x) = ke^{cx^2/2}$, $c<0$: normal with mean 0.

6.3 Sums of independent random variables

Motivation. Sums of independent variables have convolution laws; normal, Poisson and gamma families are closed under sums.

For independent X, Y with densities,

$$F_{X+Y}(a) = \int F_X(a-y)f_Y(y)\,dy, \qquad f_{X+Y}(a) = \int f_X(a-y)f_Y(y)\,dy,$$

the **convolution** of f_X and f_Y.

Example 6.8 (Sums of uniforms)

For i.i.d. $U(0,1)$, $f_{X+Y}(a) = a$ on $[0,1]$, $2-a$ on $[1,2]$; by induction $P(S_n \le x) = x^n/n!$ on $[0,1]$. With $N = \min\{n : S_n > 1\}$, $P(N > n) = 1/n!$, so

$$E[N] = \sum_{n\ge 0} P(N>n) = \sum_{n\ge 0}\frac{1}{n!} = e.$$

Proposition 6.2

For independent variables:
(a) $\text{Gamma}(s,\lambda)+\text{Gamma}(t,\lambda)\sim\text{Gamma}(s+t,\lambda)$; in particular a sum of n independent $\text{Exp}(\lambda)$ is $\text{Gamma}(n,\lambda)$ and $Z_1^2+\cdots+Z_n^2\sim\chi^2_n$.
(b) $\sum N(\mu_i,\sigma_i^2)\sim N(\sum\mu_i,\sum\sigma_i^2)$.
(c) $\text{Poisson}(\lambda_1)+\text{Poisson}(\lambda_2)\sim\text{Poisson}(\lambda_1+\lambda_2)$.
(d) $\text{Bin}(n,p)+\text{Bin}(m,p)\sim\text{Bin}(n+m,p)$.

Proof. (a) The convolution is $\frac{\lambda e^{-\lambda a}}{\Gamma(s)\Gamma(t)}\lambda^{s+t-1}\int_0^a y^{t-1}(a-y)^{s-1}\,dy = Ca^{s+t-1}e^{-\lambda a}$ $(y=ax)$, a gamma density; C is forced.

(b) Reduce to $N(0,\sigma^2)+N(0,1)$ and complete the square (or use mgfs, Lecture 7).

(c) $P(X+Y=n)=\sum_k e^{-\lambda_1}\frac{\lambda_1^k}{k!}e^{-\lambda_2}\frac{\lambda_2^{n-k}}{(n-k)!}=e^{-(\lambda_1+\lambda_2)}\frac{(\lambda_1+\lambda_2)^n}{n!}$ by the binomial theorem.

(d) Count successes in $n+m$ trials, or use Vandermonde. ∎

Example 6.9

26 games won w.p. 0.4 and 18 w.p. 0.7, independently: wins have mean 23, variance 10.02, and $P(\text{wins}\ge 25)\approx 1-\Phi(1.5/\sqrt{10.02})\approx 0.32$.

6.4 Conditional distributions

Discrete: $p_{X|Y}(x\mid y)=p(x,y)/p_Y(y)$. Jointly continuous:

$$f_{X|Y}(x\mid y)=\frac{f(x,y)}{f_Y(y)},\qquad P(X\in A\mid Y=y)=\int_A f_{X|Y}(x\mid y)\,dx,$$

(from $P(x<X<x+dx\mid y<Y<y+dy)\approx f(x,y)\,dx\,dy/(f_Y(y)\,dy)$). Independence: the conditional law is free of y.

Example 6.10

(a) Independent Poissons λ_1,λ_2: given $X+Y=n$, $X\sim\text{Bin}(n,\lambda_1/(\lambda_1+\lambda_2))$. Independent $\text{Bin}(n_1,p)$, $\text{Bin}(n_2,p)$: given the sum, X is hypergeometric, free of p.
(b) Multinomial given $X_r=n_r$: multinomial with $n-n_r$ trials, probabilities $p_i/(1-p_r)$.
(c) $f(x,y)=e^{-x/y}e^{-y}/y$: $f_{X|Y}(x\mid y)=\frac{1}{y}e^{-x/y}$, so $P(X>1\mid Y=y)=e^{-1/y}$.

Example 6.11 (The bivariate normal)

(X,Y) is bivariate normal with parameters $\mu_x,\mu_y,\sigma_x,\sigma_y,\rho$ $(|\rho|<1)$ if

$$f(x,y)=\frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}\exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[u^2-2\rho uv+v^2\Big]\Big\},\quad u=\frac{x-\mu_x}{\sigma_x},\ v=\frac{y-\mu_y}{\sigma_y}.$$

Completing the square in u: given $Y=y$, X is normal with

$$E[X\mid Y=y]=\mu_x+\rho\frac{\sigma_x}{\sigma_y}(y-\mu_y),\qquad \text{Var}(X\mid Y=y)=\sigma_x^2(1-\rho^2),$$

and $Y\sim N(\mu_y,\sigma_y^2)$. Independence iff $\rho=0$ (ρ is the correlation, Lecture 7).

Example 6.12 (Mixed variables: beta–binomial)

$P \sim U(0,1)$, $N \mid P = p \sim \mathrm{Bin}(n,p)$: $P(N=k) = \binom{n}{k} B(k+1, n-k+1) = \frac{1}{n+1}$, uniform on $\{0,\dots,n\}$, and $P \mid N = k \sim \mathrm{Beta}(k+1, n-k+1)$ (cf. Example 3.12).

6.5 Order statistics

For i.i.d. $X_1,\dots,X_n$ with density f, cdf F, the $n!$ assignments to the ordered values $X_{(1)} \le \cdots \le X_{(n)}$ are equally likely:

$$f_{X_{(1)},\dots,X_{(n)}}(x_1,\dots,x_n) = n!\, f(x_1)\cdots f(x_n), \qquad x_1 < \cdots < x_n.$$

$X_{(j)}$ is near x when $j-1$ observations are below, one near, $n-j$ above:

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!\,(n-j)!} F(x)^{j-1}(1-F(x))^{n-j} f(x).$$

Also $P(X_{(j)} \le x) = \sum_{k=j}^{n} \binom{n}{k} F(x)^k (1-F(x))^{n-k}$. For uniforms $X_{(j)} \sim \mathrm{Beta}(j, n-j+1)$, $E[X_{(j)}] = j/(n+1)$.

Example 6.13

(a) Three uniform points on $[0,1]$; probability no two within $d \le \frac{1}{2}$:

$$3! \int_{x_1 < x_2 - d,\ x_2 < x_3 - d} dx_1\, dx_2\, dx_3 = (1-2d)^3,$$

($y_1 = x_1$, $y_2 = x_2 - d$, $y_3 = x_3 - 2d$ maps onto the ordered simplex in $[0, 1-2d]^3$).
(b) The median of $2n+1$ uniforms has density $\frac{(2n+1)!}{n!\,n!} x^n (1-x)^n$; the range $R = X_{(n)} - X_{(1)}$ has $P(R \le a) = n \int f(x)\{F(x+a) - F(x)\}^{n-1}\, dx$, for uniforms $f_R(a) = n(n-1)a^{n-2}(1-a)$.

6.6 Functions of random variables: the Jacobian method

Motivation. The Jacobian changes variables in joint densities as in double integrals.

Theorem 6.3

Let (X_1, X_2) have density f, and let $y_1 = g_1(x_1,x_2)$, $y_2 = g_2(x_1,x_2)$ be a one-to-one differentiable map with inverse $x_i = h_i(y_1,y_2)$ and nonzero Jacobian $J(x_1,x_2) = \det(\partial g_i/\partial x_j)$. Then $Y_i = g_i(X_1,X_2)$ have joint density

$$f_{Y_1,Y_2}(y_1,y_2) = f\big(h_1(y),h_2(y)\big)\,|J(h_1(y),h_2(y))|^{-1}.$$

(The change of variables formula; also in n dimensions.)

Example 6.14 (Polar coordinates of normals)

X, Y i.i.d. $N(0,1)$, $R = \sqrt{X^2+Y^2}$, $\Theta \in [0, 2\pi)$; $(x,y) \mapsto (r,\theta)$ has Jacobian $1/r$:

$$f_{R,\Theta}(r,\theta) = \frac{1}{2\pi} e^{-r^2/2}\, r = \frac{1}{2\pi} \cdot r e^{-r^2/2}:$$

$R \perp \Theta$, Θ uniform, $R^2 \sim \mathrm{Exp}(1/2) = \chi^2_2$. Reversed (Box–Muller): $X =$

$\sqrt{-2\log U_1}\cos(2\pi U_2)$, $Y = \sqrt{-2\log U_1}\sin(2\pi U_2)$ are i.i.d. $N(0,1)$.

Example 6.15 (Gamma and beta)

Independent $X \sim \text{Gamma}(\alpha,\lambda)$, $Y \sim \text{Gamma}(\beta,\lambda)$, $U = X+Y$, $V = X/(X+Y)$: $x = uv$, $y = u(1-v)$, $|J|^{-1} = u$,

$$f_{U,V}(u,v) = \frac{\lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} v^{\alpha-1}(1-v)^{\beta-1}.$$

So $U \sim \text{Gamma}(\alpha+\beta,\lambda)$ and $V \sim \text{Beta}(\alpha,\beta)$ are independent, and $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$. (E.g. for two exponential jobs in sequence, the first's share of time is uniform and independent of the total.)

Example 6.16 (Spacings of exponentials)

For i.i.d. $\text{Exp}(\lambda)$, the spacings $Y_1 = X_{(1)}$, $Y_j = X_{(j)} - X_{(j-1)}$ are independent, $Y_j \sim \text{Exp}((n-j+1)\lambda)$: the map has Jacobian 1 and $\sum_j x_{(j)} = \sum_j (n-j+1)y_j$, so $n!\lambda^n e^{-\lambda\sum x_{(j)}}$ factors (survivors restart by memorylessness). Hence $E[X_{(n)}] = \frac{1}{\lambda}\sum_{k=1}^{n}\frac{1}{k}$.

6.7 Exchangeable random variables

$X_1,\dots,X_n$ are **exchangeable** if every permutation has the same joint law. Besides i.i.d. variables, e.g. the indicators that the i-th ball drawn without replacement is white: a sequence with k whites has probability $\frac{m(m-1)\cdots(m-k+1)\,(N-m)\cdots(N-m-n+k+1)}{N(N-1)\cdots(N-n+1)}$, depending only on k; so each draw is white w.p. m/N (Example 2.1(b)). Pólya's urn also gives exchangeable draws.

6.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): integrating out, factorizing survival functions, and Bayes's rule for densities.

Exercise 6.5 (Ratio and product of independent variables)

Let X and Y be independent positive continuous random variables with densities f_X and f_Y.
(a) Show that $Z = X/Y$ has density $f_Z(z) = \int_0^\infty y\, f_X(zy) f_Y(y)\, dy$, $z > 0$.
(b) Show that $W = XY$ has density $f_W(w) = \int_0^\infty \frac{1}{y} f_X(w/y) f_Y(y)\, dy$, $w > 0$.
(c) Evaluate f_Z when $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(\mu)$.

Solution.

(a)

Step 1. The joint density is $f_X(x)f_Y(y)$ (Proposition 6.1); for $z > 0$

$$F_Z(z) = P(X \le zY) = \int_0^\infty \Big(\int_0^{zy} f_X(x)\,dx\Big) f_Y(y)\,dy.$$

Step 2. Substitute $x = uy$ ($dx = y\,du$) and exchange the order (Tonelli):

$$F_Z(z) = \int_0^z \Big(\int_0^\infty y\, f_X(uy) f_Y(y)\,dy\Big)\,du.$$

So the bracket is a density of Z.

(b) Likewise $F_W(w) = \int_0^\infty \int_0^{w/y} f_X(x)\,dx\, f_Y(y)\,dy$; substitute $x = u/y$ and exchange: $F_W(w) = \int_0^w \int_0^\infty \frac{1}{y} f_X(u/y) f_Y(y)\,dy\,du$.

(c) By (a), for $z > 0$,

$$f_Z(z) = \int_0^\infty y\,\lambda e^{-\lambda z y} \mu e^{-\mu y}\,dy = \frac{\lambda\mu}{(\lambda z + \mu)^2},$$

($\int_0^\infty y e^{-ay}\,dy = 1/a^2$, $a = \lambda z + \mu$). ■

Exercise 6.8 (Hazard rate of a minimum)

Let X and Y be independent positive continuous random variables with hazard rates λ_X and λ_Y, and let $W = \min(X, Y)$.
(a) Express the distribution function of W through F_X and F_Y.
(b) Show that $\lambda_W(t) = \lambda_X(t) + \lambda_Y(t)$.

Solution.

(a) $W > t$ iff $X > t$ and $Y > t$: $\bar{F}_W = \bar{F}_X \bar{F}_Y$, $F_W(t) = 1 - \{1 - F_X(t)\}\{1 - F_Y(t)\}$.

(b)

Step 1. Where f_X, f_Y are continuous, by the product rule

$$f_W(t) = -\bar{F}_W'(t) = f_X(t)\bar{F}_Y(t) + \bar{F}_X(t) f_Y(t).$$

Step 2. Divide by $\bar{F}_W(t) = \bar{F}_X(t)\bar{F}_Y(t) > 0$.

So $\min(\mathrm{Exp}(\lambda), \mathrm{Exp}(\mu)) \sim \mathrm{Exp}(\lambda + \mu)$: competing risks add. ■

Exercise 6.14 (Two geometric waiting times given their sum)

Let X and Y be independent geometric random variables with the same parameter p. Find $P(X = i \mid X + Y = n)$.

Solution.

Step 1. (Guess.) X and $X + Y$ are the times of the first and second heads of a p-coin; given the second at flip n, the first should be uniform on $1, \ldots, n-1$.

Step 2. (Verification.) With $q = 1 - p$, $n \geq 2$, $1 \leq i \leq n - 1$,

$$P(X = i,\, Y = n - i) = q^{i-1}p \cdot q^{n-i-1}p = p^2 q^{n-2},$$

free of i; so $P(X + Y = n) = (n-1)p^2 q^{n-2}$ and

$$P(X = i \mid X + Y = n) = \frac{1}{n-1}, \qquad i = 1, \ldots, n-1.$$

■

Exercise 6.21 (Poisson counts with a gamma rate)

The moisture W on a day is $\mathrm{Gamma}(t, \lambda)$, and given $W = w$ the number N of accidents that day is $\mathrm{Poisson}(w)$. Show that given $N = n$, W is $\mathrm{Gamma}(t + n, \lambda + 1)$.

Solution.

Step 1. By Bayes's formula for the mixed pair (as in Example 6.12),

$$f_{W|N}(w \mid n) = \frac{P(N = n \mid W = w) f_W(w)}{P(N = n)}, \qquad P(N = n) = \int_0^\infty P(N = n \mid W = w) f_W(w)\, dw.$$

Step 2. For $w > 0$,

$$P(N = n \mid W = w) f_W(w) = e^{-w} \frac{w^n}{n!} \cdot \frac{\lambda e^{-\lambda w} (\lambda w)^{t-1}}{\Gamma(t)} = C_n\, e^{-(\lambda+1)w} w^{t+n-1},$$

with $C_n = \lambda^t / \{n!\, \Gamma(t)\}$ free of w.

Step 3. So $f_{W|N}(w \mid n) \propto e^{-(\lambda+1)w} w^{t+n-1}$, the Gamma$(t + n, \lambda + 1)$ kernel; proportional densities are equal. In particular $E[W \mid N = n] = (t + n)/(\lambda + 1)$. ■

Lecture 7. Properties of Expectation

7.1 Expectation of sums

$E[g(X,Y)] = \sum_x \sum_y g(x,y)p(x,y)$ or $\iint g(x,y)f(x,y)\,dx\,dy$ (as Proposition 5.2). With $g = x + y$: $E[\sum X_i] = \sum E[X_i]$, no independence needed. $X \geq Y$ implies $E[X] \geq E[Y]$.

Example 7.1

(a) Accident at $X \sim U(0,L)$, ambulance at independent $Y \sim U(0,L)$: $E|X - Y| = \frac{1}{L^2}\int_0^L \int_0^L |x-y|\,dx\,dy = \frac{L}{3}$.
(b) Boole: with X the number of A_i occurring and $Y = \mathbb{1}\{X \geq 1\} \leq X$, $P(\bigcup A_i) = E[Y] \leq \sum P(A_i)$.
(c) $E[\bar{X}] = \mu$ for identically distributed X_i.

Example 7.2 (Indicator calculations)

(a) Matches: $E[\#\text{matches}] = N \cdot \frac{1}{N} = 1$ for every N.
(b) Coupons: with i types held, the wait for a new one is geometric with parameter $(N-i)/N$, so

$$E[T] = \sum_{i=0}^{N-1} \frac{N}{N-i} = N\Big(1 + \frac{1}{2} + \cdots + \frac{1}{N}\Big) \approx N \log N.$$

Distinct types in n purchases: $N\{1 - (1 - 1/N)^n\}$ in expectation.
(c) Runs: in a random arrangement of n ones and m zeros, with I_i indicating a run of ones starting at i, $P(I_1 = 1) = \frac{n}{n+m}$ and $P(I_i = 1) = \frac{mn}{(n+m)(n+m-1)}$ $(i > 1)$, so the expected number of runs is $\frac{n(m+1)}{n+m}$.
(d) Quicksort (random pivot, compare, recurse): the i-th and j-th smallest are compared iff one of them is the first pivot among the $j-i+1$ values between them, probability $2/(j-i+1)$:

$$E[\#\text{comparisons}] = \sum_{i<j} \frac{2}{j-i+1} = 2(n+1)\sum_{k=1}^{n} \frac{1}{k} - 4n \approx 2n \log n.$$

Proposition 7.1 (Moments of the number of events)

If X is the number of the events $A_1, \ldots, A_n$ that occur, then

$$E\Big[\binom{X}{k}\Big] = \sum_{i_1 < \cdots < i_k} P(A_{i_1} \cdots A_{i_k}).$$

Proof. $\binom{X}{k} = \sum_{i_1<\cdots<i_k} I_{i_1} \cdots I_{i_k}$. ∎

$k = 2$: $E[X^2] = E[X] + 2\sum_{i<j} P(A_iA_j)$. For matches $P(A_iA_j) = \frac{1}{N(N-1)}$, so $E[X(X-1)] = 1$ and $\mathrm{Var}(X) = 1$, as for Poisson(1).

7.2 Covariance and variance of sums

Motivation. The variance of a sum expands into covariances.

Proposition 7.2

If X, Y are independent, $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

Proof. $\iint g(x)h(y)f_X(x)f_Y(y)\,dx\,dy$ factors. ■

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$. Independence implies Cov $= 0$, not conversely: X uniform on $\{-1,0,1\}$, $Y = X^2$ give Cov $= E[X^3] = 0$.

Proposition 7.3

Covariance is symmetric and bilinear: $\mathrm{Cov}(\sum_i a_iX_i, \sum_j b_jY_j) = \sum_{i,j} a_ib_j \mathrm{Cov}(X_i, Y_j)$, and $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$. Hence

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j),$$

which is $\sum \mathrm{Var}(X_i)$ for pairwise independent variables.

Example 7.3

(a) For i.i.d. X_i, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, and $\sum(X_i - \bar{X})^2 = \sum(X_i - \mu)^2 - n(\bar{X} - \mu)^2$ gives

$$E[S^2] = \frac{1}{n-1}(n\sigma^2 - n\cdot\sigma^2/n) = \sigma^2, \qquad S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2.$$

(b) Hypergeometric: for draw indicators, $\mathrm{Cov}(X_i, X_j) = \frac{m(m-1)}{N(N-1)} - \frac{m^2}{N^2} = -\frac{m(N-m)}{N^2(N-1)}$, giving $\mathrm{Var}(X) = n\frac{m}{N}(1 - \frac{m}{N})\frac{N-n}{N-1}$.
(c) Multinomial: $N_i = \sum_k I_i(k)$ with $I_i(k)I_j(k) = 0$ and independent trials, so $\mathrm{Cov}(N_i, N_j) = n(0 - p_ip_j) = -np_ip_j$.

The **correlation** is $\rho(X,Y) = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}$.

Proposition 7.4

$-1 \le \rho(X,Y) \le 1$, with $\rho = \pm 1$ iff $Y = a + bX$ with probability 1 for some b of the same sign.

Proof. $0 \le \mathrm{Var}(X/\sigma_X \pm Y/\sigma_Y) = 2(1 \pm \rho)$, with equality iff $X/\sigma_X \pm Y/\sigma_Y$ is constant. ■

ρ measures *linear* dependence; for the bivariate normal (Example 6.11) $\rho(X,Y) = \rho$.

7.3 Conditional expectation

Motivation. $E[X \mid Y]$, the best guess of X given Y, averages back to $E[X]$: compute expectations by conditioning.

$E[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y)$ or $\int x f_{X|Y}(x \mid y)\,dx$; $E[X \mid Y] = g(Y)$ with $g(y) = E[X \mid Y = y]$.

Theorem 7.5 (Tower property)

$E[X] = E[E[X \mid Y]]$, i.e. $E[X] = \sum_y E[X \mid Y = y]P(Y = y)$ or $\int E[X \mid Y = y]f_Y(y)\,dy$.

Proof. Discrete case: $\sum_y \sum_x x p_{X|Y}(x \mid y) p_Y(y) = \sum_x \sum_y x p(x,y) = E[X]$. ∎

Example 7.4 (The trapped miner)

A miner picks one of three doors at random: safety in 3 hours, or back to the mine after 5 or 7 hours. Conditioning on the first door, $E[X] = \frac{1}{3}\{3 + (5 + E[X]) + (7 + E[X])\}$, so $E[X] = 15$.

Example 7.5 (Random sums)

If $N \geq 0$ is integer-valued, independent of i.i.d. X_i, then $E[\sum_{i=1}^{N} X_i \mid N = n] = nE[X]$, so

$$E\Big[\sum_{i=1}^{N} X_i\Big] = E[N]E[X].$$

Example 7.6 (Waiting for k consecutive successes)

N_k: trials until k consecutive successes. After N_{k-1}, one more trial succeeds or restarts: $E[N_k \mid N_{k-1}] = N_{k-1} + 1 + (1-p)E[N_k]$, giving $E[N_k] = \frac{1}{p} + \frac{E[N_{k-1}]}{p}$ and

$$E[N_k] = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}.$$

For a fair coin, $E[N_2] = 6$ and $E[N_3] = 14$.

Probabilities by conditioning. $P(E) = E[P(E \mid Y)]$; e.g. $P(X < Y) = \int F_X(y) f_Y(y)\, dy$ for independent continuous X, Y, and $P(X + Y < a) = \int F_X(a - y) f_Y(y)\, dy$.

Example 7.7 (The best-prize problem)

n distinct prizes appear in random order; each is accepted or rejected on the spot, knowing only relative ranks. Reject the first k, then accept the first prize better than all earlier. If the best is at $X = i > k$, we win iff the best of the first $i - 1$ is among the first k, probability $k/(i-1)$:

$$P_k(\text{best}) = \frac{1}{n} \sum_{i=k+1}^{n} \frac{k}{i-1} \approx \frac{k}{n} \log \frac{n}{k}.$$

Maximizing $x \log(1/x)$, $x = k/n$, gives $k \approx n/e$ and success probability $\to 1/e \approx 0.368$.

Example 7.8 (Uniform sums revisited)

Let $m(x) = E[N(x)]$, $N(x) = \min\{n : U_1 + \cdots + U_n > x\}$, $0 \leq x \leq 1$. Conditioning on $U_1 = y$ ($N = 1$ if $y > x$, else $1 + N'(x - y)$): $m(x) = 1 + \int_0^x m(u)\, du$, so $m' = m$, $m(0) = 1$, $m(x) = e^x$ (cf. Example 6.8).

Proposition 7.6 (Conditional variance formula)

With $\mathrm{Var}(X \mid Y) = E[(X - E[X \mid Y])^2 \mid Y]$,

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y]).$$

Proof. $E[\mathrm{Var}(X \mid Y)] = E[X^2] - E[(E[X \mid Y])^2]$ and $\mathrm{Var}(E[X \mid Y]) = E[(E[X \mid Y])^2] - (E[X])^2$; add. ∎

Example 7.9

(a) Random sum $S = \sum_1^N X_i$: $E[S \mid N] = N\mu$, $\mathrm{Var}(S \mid N) = N\sigma^2$, so $\mathrm{Var}(S) = E[N]\sigma^2 + \mu^2 \mathrm{Var}(N)$; for $N \sim \mathrm{Poisson}(\lambda)$, $\mathrm{Var}(S) = \lambda E[X^2]$.
(b) Geometric: with Y indicating first-trial success, $E[N \mid Y] \in \{1, 1 + 1/p\}$, $\mathrm{Var}(N \mid Y) \in \{0, \mathrm{Var}(N)\}$; solving, $\mathrm{Var}(N) = (1-p)/p^2$.

7.4 Conditional expectation and prediction

Motivation. $E[X \mid Y]$ minimizes mean squared prediction error; restricting to linear predictors gives regression.

Proposition 7.7

$E[(Y - g(X))^2] \geq E[(Y - E[Y \mid X])^2]$ for every function g: $E[Y \mid X]$ is the best predictor of Y from X in mean square error.

Proof. $E[(Y-c)^2 \mid X]$ is minimized by $c = E[Y \mid X]$; take expectations. ∎

Minimizing $E[(Y - a - bX)^2]$ over linear predictors,

$$b = \rho \frac{\sigma_Y}{\sigma_X}, \qquad a = \mu_Y - b\mu_X, \qquad \mathrm{MSE} = \sigma_Y^2(1-\rho^2).$$

For the bivariate normal $E[Y \mid X]$ is linear. Galton's "regression to the mean" is the factor $|\rho| < 1$.

7.5 Moment generating functions

Motivation. The mgf turns independent sums into products and determines the distribution.

$M(t) = E[e^{tX}]$; when finite near 0, $M^{(n)}(0) = E[X^n]$.

Distribution	$M(t)$	valid for
$\mathrm{Bin}(n,p)$	$(pe^t + 1 - p)^n$	all t
$\mathrm{Poisson}(\lambda)$	$\exp\{\lambda(e^t - 1)\}$	all t
$\mathrm{Geometric}(p)$	$pe^t/\{1-(1-p)e^t\}$	$(1-p)e^t < 1$
$\mathrm{Exp}(\lambda)$	$\lambda/(\lambda - t)$	$t < \lambda$
$\mathrm{Gamma}(\alpha, \lambda)$	$\{\lambda/(\lambda-t)\}^\alpha$	$t < \lambda$
$N(\mu, \sigma^2)$	$\exp\{\mu t + \sigma^2 t^2/2\}$	all t

Normal: $E[e^{tZ}] = \frac{1}{\sqrt{2\pi}} \int e^{-(x-t)^2/2 + t^2/2}\, dx = e^{t^2/2}$, and $X = \mu + \sigma Z$.

Theorem 7.8

(a) If X, Y are independent, $M_{X+Y}(t) = M_X(t)M_Y(t)$.
(b) (Uniqueness) If $M_X = M_Y$ on a neighbourhood of 0, then X and Y have the same distribution.

(a) is Proposition 7.2 with $g = e^{tx}$, $h = e^{ty}$;

(b) is quoted. Together they give Proposition 6.2, e.g. $e^{\mu_1 t+\sigma_1^2 t^2/2}e^{\mu_2 t+\sigma_2^2 t^2/2}$ is the $N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$ mgf.

Example 7.10

(a) $E[e^{tZ^2}] = (1-2t)^{-1/2}$, so $\sum_1^n Z_i^2$ has the Gamma$(n/2, 1/2)$ mgf $(1-2t)^{-n/2}$.
(b) Random sums: $M_S(t) = E[M_X(t)^N]$; differentiating at 0 recovers Example 7.9.
(c) $M(t) = e^{3(e^t-1)}$: $X \sim$ Poisson(3), $P(X=0) = e^{-3}$.

The **joint mgf** $E[e^{t_1X_1+\cdots+t_nX_n}]$ determines the joint law, and factors iff the X_i are independent.

Example 7.11

For i.i.d. $N(\mu, \sigma^2)$ X, Y, $X+Y \perp X-Y$: $t(X+Y)+s(X-Y) = (t+s)X+(t-s)Y$, so

$$E[e^{t(X+Y)+s(X-Y)}] = e^{2\mu t+\sigma^2(t^2+s^2)} = E[e^{t(X+Y)}]\,E[e^{s(X-Y)}].$$

7.6 The multivariate normal distribution

If $Z_1, \ldots, Z_n$ are i.i.d. $N(0,1)$ and $X_i = \mu_i + \sum_j a_{ij}Z_j$, $(X_1, \ldots, X_m)$ is **multivariate normal**; every $\sum t_iX_i$ is normal with mean $\sum t_i\mu_i$ and variance $\sum_{i,j} t_it_j \operatorname{Cov}(X_i, X_j)$, so

$$M(t_1, \ldots, t_m) = \exp\Big\{\sum_i t_i\mu_i + \frac{1}{2}\sum_{i,j} t_it_j \operatorname{Cov}(X_i, X_j)\Big\}:$$

means and covariances determine the law, and uncorrelated components are independent.

Theorem 7.9 (Sample mean and variance)

If $X_1, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\bar{X}$ and S^2 are independent, $\bar{X} \sim N(\mu, \sigma^2/n)$, and $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Proof. $(\bar{X}, X_1-\bar{X}, \ldots, X_n-\bar{X})$ is multivariate normal with $\operatorname{Cov}(\bar{X}, X_i-\bar{X}) = 0$, so $\bar{X}$ is independent of the deviations and of S^2. Next,

$$\sum_{i=1}^n \Big(\frac{X_i-\mu}{\sigma}\Big)^2 = \frac{(n-1)S^2}{\sigma^2} + \Big(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\Big)^2.$$

The left side is χ^2_n, the last term χ^2_1 independent of the first; dividing mgfs, $(n-1)S^2/\sigma^2$ has mgf $(1-2t)^{-(n-1)/2}$. ∎

7.7 General definition of expectation

For general $X \geq 0$, $E[X] = \lim E[X_n]$ for simple $X_n \uparrow X$ (Lebesgue integral), or $\int_0^\infty P(X > x)\,dx$; $E[X] = E[X^+] - E[X^-]$ in general. All properties persist; e.g. if $X = Y$ w.p. p and W otherwise, $E[X] = pE[Y] + (1-p)E[W]$.

7.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): indicators, conditioning, and the moments of a branching process.

Exercise 7.13 (Record values)

Let $X_1, \dots, X_n$ be i.i.d. continuous random variables. A record occurs at time j if $X_j \geq X_i$ for all $i \leq j$. Show that
(a) $E[\text{number of records}] = \sum_{j=1}^{n} 1/j$;
(b) $\operatorname{Var}(\text{number of records}) = \sum_{j=1}^{n} (j-1)/j^2$.

Solution.

Step 1. (Ranks are uniform.) $P(X_i = X_k) = 0$ for $i \neq k$, and the joint density of $(X_1, \dots, X_m)$ is symmetric, so each of the $m!$ orderings has probability $1/m!$.

Step 2. I_j indicates $X_j = \max(X_1, \dots, X_j)$; $(j-1)!$ of $j!$ orderings put X_j on top, so $P(I_j = 1) = 1/j$, giving (a).

Step 3. For $i < j$, orderings with X_j on top correspond to the $(j-1)!$ orderings of $X_1, \dots, X_{j-1}$, a fraction $1/i$ of which puts X_i above $X_1, \dots, X_{i-1}$. So $P(I_i I_j = 1) = \frac{1}{ij} = P(I_i = 1)P(I_j = 1)$ and $\operatorname{Cov}(I_i, I_j) = 0$.

Step 4. By Proposition 7.3,

$$\operatorname{Var}\Big(\sum_{j=1}^{n} I_j\Big) = \sum_{j=1}^{n} \operatorname{Var}(I_j) = \sum_{j=1}^{n} \frac{1}{j}\Big(1 - \frac{1}{j}\Big) = \sum_{j=1}^{n} \frac{j-1}{j^2}.$$

■

Exercise 7.20 (The conditional covariance formula)

For X, Y, Z with $E[X^2], E[Y^2] < \infty$ define $\operatorname{Cov}(X, Y \mid Z) = E\big[(X - E[X \mid Z])(Y - E[Y \mid Z]) \mid Z\big]$.
(a) Show that $\operatorname{Cov}(X, Y \mid Z) = E[XY \mid Z] - E[X \mid Z]E[Y \mid Z]$.
(b) Prove that $\operatorname{Cov}(X, Y) = E[\operatorname{Cov}(X, Y \mid Z)] + \operatorname{Cov}(E[X \mid Z], E[Y \mid Z])$.
(c) Deduce the conditional variance formula.

Solution.

(a) $E[g(Z)V \mid Z] = g(Z)E[V \mid Z]$; expanding with $g(Z) = E[X \mid Z]$, $E[Y \mid Z]$,

$$\begin{aligned}\operatorname{Cov}(X, Y \mid Z) &= E[XY \mid Z] - E[X \mid Z]E[Y \mid Z] - E[Y \mid Z]E[X \mid Z] + E[X \mid Z]E[Y \mid Z] \\ &= E[XY \mid Z] - E[X \mid Z]E[Y \mid Z].\end{aligned}$$

(b) Take expectations in (a) (Theorem 7.5):

$$E[\operatorname{Cov}(X, Y \mid Z)] = E[XY] - E\big[E[X \mid Z]E[Y \mid Z]\big].$$

and, since $E\big[E[X \mid Z]\big] = E[X]$,

$$\operatorname{Cov}(E[X \mid Z], E[Y \mid Z]) = E\big[E[X \mid Z]E[Y \mid Z]\big] - E[X]E[Y].$$

Add.

(c) $Y = X$ gives Proposition 7.6. ■

Exercise 7.35 (Generating functions as probabilities)

Let X be nonnegative integer valued with $p_j = P(X = j)$ and generating function $\phi(s) = E[s^X] = \sum_j p_j s^j$. For $0 < s < 1$ let Y be geometric with parameter $1 - s$, independent of X. Show that $\phi(s) = P(X < Y)$.

Solution.

Step 1. $P(Y = k) = s^{k-1}(1 - s)$, so $P(Y > j) = s^j$.

Step 2. Conditioning on X,

$$P(X < Y) = \sum_{j=0}^{\infty} P(X = j)P(Y > j) = \sum_{j=0}^{\infty} p_j s^j = \phi(s).$$

■

Exercise 7.44 (Moments of a branching process)

Each individual of a population produces j offspring with probability P_j, independently of all others; $\mu = \sum_j jP_j$ and $\sigma^2 = \sum_j (j - \mu)^2 P_j < \infty$. Let X_n be the size of generation n, with $X_0 = 1$. Show that
(a) $E[X_n] = \mu E[X_{n-1}]$, hence $E[X_n] = \mu^n$;
(b) $\operatorname{Var}(X_n) = \sigma^2 \mu^{n-1} + \mu^2 \operatorname{Var}(X_{n-1})$;
(c) $\operatorname{Var}(X_n) = \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}$ if $\mu \neq 1$, and $\operatorname{Var}(X_n) = n\sigma^2$ if $\mu = 1$.

Solution.

Step 1. $X_n = \sum_{i=1}^{X_{n-1}} Z_i$ with Z_i the offspring counts, i.i.d. (mean μ, variance σ^2) and independent of X_{n-1}:

$$E[X_n \mid X_{n-1}] = \mu X_{n-1}, \qquad \operatorname{Var}(X_n \mid X_{n-1}) = \sigma^2 X_{n-1}.$$

(a) Tower property and $E[X_0] = 1$.

(b) Proposition 7.6 and (a):

$$\operatorname{Var}(X_n) = E[\sigma^2 X_{n-1}] + \operatorname{Var}(\mu X_{n-1}) = \sigma^2 \mu^{n-1} + \mu^2 \operatorname{Var}(X_{n-1}).$$

(c) $V_n = \operatorname{Var}(X_n)$, $V_0 = 0$. If $\mu = 1$, $V_n = \sigma^2 + V_{n-1} = n\sigma^2$. If $\mu \neq 1$, by induction

$$\begin{aligned} V_n &= \sigma^2 \mu^{n-1} + \mu^2 \sigma^2 \mu^{n-2} \frac{\mu^{n-1} - 1}{\mu - 1} = \sigma^2 \mu^{n-1} \frac{(\mu - 1) + \mu(\mu^{n-1} - 1)}{\mu - 1} \\ &= \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}. \end{aligned}$$

■

Lecture 8. Limit Theorems

8.1 Markov and Chebyshev inequalities; the weak law

Motivation. Markov and Chebyshev bound tails from a mean or variance alone, and give the weak law.

Proposition 8.1 (Markov's inequality)

If $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq E[X]/a$.

Proof. $a\mathbb{1}\{X \geq a\} \leq X$; take expectations. ■

Proposition 8.2 (Chebyshev's inequality)

If X has mean μ and variance σ^2, then $P(|X - \mu| \geq k) \leq \sigma^2/k^2$ for $k > 0$.

Proof. Apply Markov's inequality to $(X - \mu)^2$ and $a = k^2$. ■

Crude, but valid for every distribution.

Example 8.1

(a) Output with mean 50: $P(X \geq 75) \leq 2/3$; with $\mathrm{Var}(X) = 25$ also $P(40 < X < 60) \geq 3/4$.
(b) $X \sim U(0, 10)$: Chebyshev gives $P(|X - 5| > 4) \leq (25/3)/16 \approx 0.52$; truly 0.2.
(c) $\mathrm{Var}(X) = 0$ gives $P(|X - \mu| > 1/n) = 0$ for all n, so $P(X = \mu) = 1$.

Theorem 8.3 (Weak law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ, then for every $\epsilon > 0$, $P(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ as $n \to \infty$.

Proof. (Finite σ^2.) $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$; Chebyshev gives $\sigma^2/(n\epsilon^2) \to 0$. ■

8.2 The central limit theorem

Motivation. Fluctuations of averages are approximately normal, whatever the distribution.

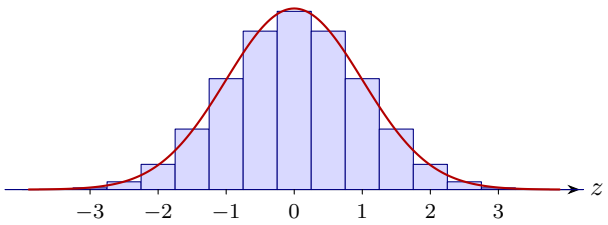


Figure. $(S_{16} - 8)/2$ *for* $S_{16} \sim \mathrm{Bin}(16, \frac{1}{2})$ *(bars, unit area) against the* $N(0, 1)$ *density.*

Theorem 8.4 (Central limit theorem)

If $X_1, X_2, \dots$ are i.i.d. with mean μ and variance $\sigma^2 \in (0,\infty)$, then for every a,

$$P\Big(\frac{X_1+\cdots+X_n-n\mu}{\sigma\sqrt{n}} \le a\Big) \to \Phi(a) \qquad (n\to\infty).$$

Key tool (quoted): $M_{Z_n}(t) \to M_Z(t)$ for all t implies $F_{Z_n}(a) \to F_Z(a)$ at continuity points.

Proof sketch. Take $\mu = 0$, $\sigma = 1$, mgf M near 0, $L = \log M$: $L(0) = 0$, $L'(0) = 0$, $L''(0) = 1$. $Z_n = \sum X_i/\sqrt{n}$ has mgf $M(t/\sqrt{n})^n$, and by L'Hôpital twice ($x = 1/\sqrt{n}$)

$$nL\Big(\frac{t}{\sqrt{n}}\Big) = \frac{L(tx)}{x^2} \to \frac{t^2 L''(0)}{2} = \frac{t^2}{2}.$$

So $M_{Z_n}(t) \to e^{t^2/2}$. (In general use characteristic functions.) ■

Hence sums of many small effects are approximately normal. The CLT also holds for independent uniformly bounded X_i with $\sum \mathrm{Var}(X_i) = \infty$: $(\sum(X_i-\mu_i))/\sqrt{\sum \sigma_i^2} \approx N(0,1)$.

Example 8.2 (How many measurements?)

Averaging n measurements of variance 4, to be 95% sure of accuracy 0.5: CLT needs $0.25\sqrt{n} \ge 1.96$, $n \ge 62$; Chebyshev needs $4/(0.25n) \le 0.05$, $n \ge 320$ (valid for any distribution).

Example 8.3

(a) 10 fair dice: mean 35, variance 350/12; with continuity correction $P(30 \le S \le 40) \approx 2\Phi(5.5/5.40) - 1 \approx 0.69$.
(b) 10 uniforms: mean 5, variance 10/12, $P(\sum X_i > 6) \approx 1 - \Phi(1.10) \approx 0.137$.
(c) Grading times with mean 20, SD 4 minutes: 25 exams within 450 minutes has probability $\approx \Phi(-2.5) \approx 0.006$.

8.3 The strong law of large numbers

Motivation. The strong law makes the whole sequence of averages converge almost surely, justifying frequencies.

Theorem 8.5 (Strong law of large numbers)

If $X_1, X_2, \dots$ are i.i.d. with finite mean μ, then $P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$.

Proof. Assume $K = E[X_i^4] < \infty$, $\mu = 0$. In $E[S_n^4]$ terms with an index appearing once vanish, leaving $nE[X^4] + 6\binom{n}{2}E[X_i^2X_j^2] \le nK + 3n(n-1)K$. So $E[\sum_n S_n^4/n^4] \le \sum_n (K/n^3 + 3K/n^2) < \infty$, hence $\sum_n S_n^4/n^4 < \infty$ a.s., forcing $\bar{X}_n \to 0$. ■

The weak law: for each large n, $\bar{X}_n$ is *probably* near μ; the strong law: a.s. the sequence eventually stays near μ. For $X_i = \mathbb{1}\{E \text{ on trial } i\}$, the long-run frequency of E is $P(E)$ a.s.

8.4 Other inequalities

Motivation. Chernoff bounds give exponential tails from the mgf; Jensen compares $E[g(X)]$ with $g(E[X])$.

Proposition 8.6 (One-sided Chebyshev)

If $E[X] = 0$ and $\mathrm{Var}(X) = \sigma^2$, then $P(X \geq a) \leq \sigma^2/(\sigma^2 + a^2)$ for $a > 0$.

Proof. For $b > 0$, $P(X \geq a) \leq P((X+b)^2 \geq (a+b)^2) \leq (\sigma^2 + b^2)/(a+b)^2$ by Markov; choose $b = \sigma^2/a$. ■

Proposition 8.7 (Chernoff bounds)

If $M(t) = E[e^{tX}]$, then $P(X \geq a) \leq e^{-ta}M(t)$ for all $t > 0$ and $P(X \leq a) \leq e^{-ta}M(t)$ for all $t < 0$.

Proof. For $t > 0$, $P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq E[e^{tX}]e^{-ta}$ by Markov. ■

Then minimize over t.

Example 8.4

(a) Standard normal: $e^{-ta+t^2/2}$ is minimized at $t = a$, so $P(Z \geq a) \leq e^{-a^2/2}$ for $a > 0$.
(b) Poisson(λ) and $i > \lambda$: minimizing $e^{\lambda(e^t-1)-ti}$ at $e^t = i/\lambda$ gives $P(X \geq i) \leq e^{-\lambda}(e\lambda)^i/i^i$.
(c) ± 1 fair plays: $M(t) = \cosh(t)^n \leq e^{nt^2/2}$, so $P(S_n \geq a) \leq e^{-a^2/(2n)}$ $(t = a/n)$; e.g. $P(S_{10} \geq 6) \leq 0.165$ (exact 0.0547).

Proposition 8.8 (Jensen's inequality)

If f is convex, then $E[f(X)] \geq f(E[X])$ whenever the expectations exist.

Proof. Convexity gives $f(x) \geq f(\mu) + f'(\mu)(x - \mu)$, $\mu = E[X]$; take expectations. ■

E.g. with concave utility u, $E[u(X)] \leq u(\mu)$: a sure return μ is preferred.

8.5 The Poisson approximation, quantified

For $X = \sum_{i=1}^n X_i$, independent $X_i \sim$ Bernoulli(p_i), and $Y \sim$ Poisson($\sum p_i$), coupling each X_i with a Poisson(p_i) variable agreeing w.p. $\geq 1 - p_i^2$ gives, for all A,

$$|P(X \in A) - P(Y \in A)| \leq \sum_{i=1}^n p_i^2.$$

For Bin(n, p) the bound is λp: good for small p, any n.

Example 8.5 (Birthdays, again)

With N the number of arrivals until the first shared birthday, $N > n$ iff no person $j \leq n$ shares with persons $1, \ldots, j-1$ (probability $(j-1)/365$ given no earlier match): $P(N > n) = \prod_{j=1}^n (1 - \frac{j-1}{365}) \approx e^{-n(n-1)/730}$, as in Example 4.6.

8.6 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): Weierstrass via the weak law, a bound for monotone pmfs, and the binomial Chernoff bound.

Exercise 8.4 (Convergence in probability and bounded functions)

Let $Z_1, Z_2, \dots$ be random variables and c a constant such that $P(|Z_n - c| > \epsilon) \to 0$ for every $\epsilon > 0$. Show that $E[g(Z_n)] \to g(c)$ for every bounded continuous g.

Solution.

Step 1. Let $|g| \le M$, $\epsilon > 0$, and $\delta > 0$ with $|g(x) - g(c)| < \epsilon$ for $|x - c| \le \delta$.

Step 2. Splitting on $\{|Z_n - c| \le \delta\}$,

$$|E[g(Z_n)] - g(c)| \le E|g(Z_n) - g(c)| \le \epsilon + 2M\, P(|Z_n - c| > \delta).$$

Step 3. So $\limsup_n |E[g(Z_n)] - g(c)| \le \epsilon$ for every ϵ. ∎

Exercise 8.5 (Bernstein polynomials)

Let f be continuous on $[0,1]$ and

$$B_n(x) = \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k} x^k (1-x)^{n-k}.$$

Prove that $B_n(x) \to f(x)$, uniformly in $x \in [0,1]$. (This is a probabilistic proof of the Weierstrass approximation theorem.)

Solution.

Step 1. For $S_n \sim \mathrm{Bin}(n, x)$, $B_n(x) = E[f(S_n/n)]$, $E[S_n/n] = x$, $\mathrm{Var}(S_n/n) \le 1/(4n)$.

Step 2. $|f| \le M$ and f is uniformly continuous: $|f(u) - f(v)| < \epsilon$ for $|u - v| \le \delta$.

Step 3. As in Exercise 8.4 ($c = x$) with Chebyshev (Proposition 8.2),

$$|B_n(x) - f(x)| \le \epsilon + 2M\, P\Big(\Big|\frac{S_n}{n} - x\Big| > \delta\Big) \le \epsilon + \frac{2M}{4n\delta^2}.$$

Step 4. The bound is free of x, so $\limsup_n \sup_x |B_n(x) - f(x)| \le \epsilon$: uniform convergence, and polynomials are dense in $C[0,1]$. ∎

Exercise 8.6(a) (Monotone probability mass functions)

Let X take values in $\{1, 2, \dots\}$ with $P(X = k)$ nonincreasing in k. Prove that $P(X = k) \le 2E[X]/k^2$.

Solution.

Step 1. With $p_i = P(X = i)$,

$$E[X] = \sum_{i \ge 1} i p_i \ge \sum_{i=1}^{k} i p_i.$$

Step 2. $p_i \ge p_k$ for $i \le k$, so $\sum_{i=1}^{k} i p_i \ge p_k \frac{k(k+1)}{2} \ge p_k \frac{k^2}{2}$. ∎

Exercise 8.11 (The Chernoff bound for the binomial)

Let $X \sim \mathrm{Bin}(n,p)$, $0<p<1$, $q=1-p$, and $np<i<n$.
(a) Show that $\min_{t>0} e^{-ti}E[e^{tX}]$ is attained where $e^t = iq/\{(n-i)p\}$.
(b) Deduce

$$P(X \geq i) \leq \frac{n^n}{i^i(n-i)^{n-i}}\, p^i q^{n-i}.$$

Solution.

(a)

Step 1. Minimize $h(t) = -ti + n\log(pe^t+q)$ over $t>0$:

$$h'(t) = -i + \frac{npe^t}{pe^t+q}, \qquad h''(t) = \frac{npqe^t}{(pe^t+q)^2} > 0,$$

so h is strictly convex.

Step 2. $h'(t)=0$ iff $e^t = iq/\{(n-i)p\}$, positive t since $i>np$.

(b) There $pe^t+q = \frac{nq}{n-i}$ and $e^{-ti} = \left(\frac{(n-i)p}{iq}\right)^i$; by Proposition 8.7

$$P(X \geq i) \leq \Big(\frac{(n-i)p}{iq}\Big)^i \Big(\frac{nq}{n-i}\Big)^n = \frac{n^n}{i^i(n-i)^{n-i}}\, p^i q^{n-i}.$$

E.g. $n=100$, $p=\frac{1}{2}$, $i=60$: bound 0.134, exact 0.028. ■

Lecture 9. Additional Topics in Probability

9.1 The Poisson process

Motivation. Random events with independent increments and no simultaneity form a Poisson process: Poisson counts, exponential gaps.

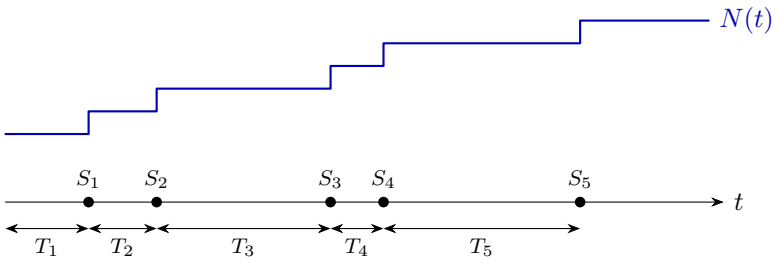


Figure. *Arrival times S_k, i.i.d.* Exp(λ) *gaps $T_k = S_k - S_{k-1}$, and the counting process $N(t) \sim$* Poisson(λt).

Definition (Poisson process)

A family $\{N(t), t \geq 0\}$ counting events in $[0, t]$ is a **Poisson process with rate** $\lambda > 0$ if (i) $N(0) = 0$;
(ii) the numbers of events in disjoint intervals are independent (independent increments);
(iii) the law of the number of events in $(s, s + t]$ depends only on t (stationary increments);
(iv) $P(N(h) = 1) = \lambda h + o(h)$ and $P(N(h) \geq 2) = o(h)$.

Proposition 9.1

For a Poisson process, $N(t) \sim$ Poisson(λt), and more generally the number of events in any interval of length t is Poisson(λt).

Proof. Split $[0, t]$ into n pieces. By (iv) and Boole, some piece has ≥ 2 events with probability $\leq n\, o(t/n) \to 0$; otherwise $N(t)$ counts pieces with an event, Bin$(n, \lambda t/n + o(t/n))$ by (ii)–(iv), with Poisson(λt) limit (Proposition 4.4). ■

Let T_1 be the first event time and T_n the n-th interarrival time.

Proposition 9.2

The interarrival times $T_1, T_2, \ldots$ are i.i.d. Exp(λ). Hence the time $S_n = T_1 + \cdots + T_n$ of the n-th event is Gamma(n, λ).

Proof. $P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$. Given $T_1 = s$, $T_2 > t$ means no event in $(s, s + t]$, probability $e^{-\lambda t}$ for every s; so $T_2 \sim$ Exp(λ) independent of T_1; induct. The gamma law: Proposition 6.2(a), or $P(S_n \leq t) = P(N(t) \geq n)$. ■

Conversely, i.i.d. exponential gaps give a Poisson process (memorylessness yields independent increments).

Example 9.1

Earthquakes at rate 2 per week: at least 3 in 2 weeks has probability $1 - 13e^{-4} \approx 0.762$; next quake more than 2 weeks away, $e^{-4} \approx 0.018$.

Example 9.2 (Location of an event)

Given $N(t) = 1$, the time T_1 of the event is uniform on $[0, t]$: for $s \le t$,

$$P(T_1 \le s \mid N(t) = 1) = \frac{P(N(s) = 1)P(N(t) - N(s) = 0)}{P(N(t) = 1)} = \frac{\lambda s e^{-\lambda s} e^{-\lambda(t-s)}}{\lambda t e^{-\lambda t}} = \frac{s}{t}.$$

Given $N(t) = n$, the event times are distributed as n ordered i.i.d. $U(0, t)$ variables.

With Example 6.4: independently marking events as type 1 with probability p gives *independent* Poisson processes of rates λp and $\lambda(1 - p)$.

9.2 Markov chains

Motivation. Memoryless evolution: n-step probabilities are matrix powers, often converging to a stationary law.

Definition (Markov chain)

A sequence $X_0, X_1, \ldots$ with values in $\{0, 1, \ldots, M\}$ is a **Markov chain** if $P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = P_{ij}$ for all states and n, with **transition probabilities** $P_{ij} \ge 0$, $\sum_j P_{ij} = 1$.

With transition matrix $\mathbf{P} = (P_{ij})$,

$$P(X_0 = i_0, \ldots, X_n = i_n) = P(X_0 = i_0) P_{i_0 i_1} \cdots P_{i_{n-1} i_n}.$$

Example 9.3

(a) Weather: rain tomorrow w.p. α after rain, β otherwise; with 0 = rain, $\mathbf{P} = \begin{pmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{pmatrix}$.
(b) Gambler's ruin: $P_{i,i+1} = p = 1 - P_{i,i-1}$ $(0 < i < N)$, $P_{00} = P_{NN} = 1$ (absorbing).
(c) Ehrenfest: M molecules in two urns, a random one switches each step; with X_n in urn 1, $P_{i,i-1} = i/M$, $P_{i,i+1} = (M - i)/M$.
(d) Weather depending on two days: (yesterday, today) is a four-state chain; enlarging the state makes longer memory Markov.

Proposition 9.3 (Chapman–Kolmogorov)

The n-step transition probabilities $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfy $P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}$; hence $(P_{ij}^{(n)}) = \mathbf{P}^n$.

Proof. Condition on the state after n steps and use the Markov property. ∎

Example 9.4

Weather with $\alpha = 0.7$, $\beta = 0.4$: $(\mathbf{P}^4)_{00} = 0.5749$, $(\mathbf{P}^4)_{10} = 0.5668$; the start is almost forgotten.

Theorem 9.4 (Limiting probabilities)

If for some n all entries of $\mathbf{P}^n$ are positive (the chain is *ergodic*), then $\pi_j = \lim_{n\to\infty} P_{ij}^{(n)}$ exists, does not depend on i, and is the unique solution of

$$\pi_j = \sum_i \pi_i P_{ij} \quad (j = 0, \dots, M), \qquad \sum_j \pi_j = 1.$$

Let $n \to \infty$ in $P_{ij}^{(n+1)} = \sum_k P_{ik}^{(n)} P_{kj}$ (existence quoted). π is **stationary** ($X_0 \sim \pi$ gives $X_n \sim \pi$), π_j is the long-run fraction of time in j, and $1/\pi_j$ the mean return time.

Example 9.5

(a) Weather: $\pi_0 = \alpha\pi_0 + \beta\pi_1$, $\pi_0 + \pi_1 = 1$ give $\pi_0 = \beta/(1+\beta-\alpha) = 4/7 \approx 0.571$ (cf. Example 9.4).
(b) Ehrenfest: $\pi_j = \binom{M}{j}2^{-M}$ satisfies $\pi_{j-1}\frac{M-j+1}{M} + \pi_{j+1}\frac{j+1}{M} = 2^{-M}\{\binom{M-1}{j-1} + \binom{M-1}{j}\} = \pi_j$, so each molecule is in either urn w.p. $\frac{1}{2}$, independently. (Period 2: $P_{ij}^{(n)}$ does not converge, but π_j is still the long-run fraction.)

9.3 Surprise, uncertainty and entropy

The surprise $S(p)$ of an event of probability p, if decreasing and continuous with $S(1) = 0$ and $S(pq) = S(p) + S(q)$, must be $-C\log_2 p$ (the equation of Proposition 5.4); take $C = 1$ (**bits**).

The **entropy** of X with probabilities $p_1, \dots, p_n$ is the expected surprise

$$H(X) = -\sum_i p_i \log_2 p_i,$$

It is 0 iff X is constant and $\le \log_2 n$, with equality iff uniform (Jensen: $\sum p_i \log(1/p_i) \le \log \sum p_i \cdot (1/p_i)$).

The joint entropy is $H(X,Y) = -\sum_{i,j} p(x_i, y_j)\log_2 p(x_i, y_j)$, and the conditional entropy $H_Y(X) = \sum_j H(X \mid Y = y_j)P(Y = y_j)$ is the uncertainty left once Y is known.

Proposition 9.5

(a) $H(X,Y) = H(Y) + H_Y(X)$.
(b) $H_Y(X) \le H(X)$, with equality iff X and Y are independent.

Proof. (a) $\log p(x_i, y_j) = \log p(y_j) + \log p(x_i \mid y_j)$; take $-E$.

(b) With $\log x \le x - 1$,

$$H_Y(X) - H(X) = \sum_{i,j} p(x_i, y_j)\log_2 \frac{p(x_i)}{p(x_i \mid y_j)} \le \frac{1}{\ln 2}\sum_{i,j} p(y_j)\{p(x_i) - p(x_i \mid y_j)\} = 0,$$

with equality iff $p(x_i \mid y_j) = p(x_i)$ whenever $p(x_i, y_j) > 0$. ∎

So observing Y never increases uncertainty on average; $H(X) - H_Y(X)$ is the information Y carries about X.

9.4 Coding theory and entropy

Transmit X (values x_i, probabilities p_i) as binary strings decodable in sequence. In a **prefix code** no codeword begins another, so concatenations decode by reading until a codeword ends.

Lemma 9.6 (Kraft's inequality)

A prefix code with codeword lengths $n_1, \dots, n_N$ exists iff $\sum_i 2^{-n_i} \le 1$.

Proof. With $L = \max n_i$, a codeword of length n_i prefixes 2^{L-n_i} strings of length L, disjointly for a prefix code: $\sum_i 2^{L-n_i} \le 2^L$. Conversely, assign codewords by increasing length, avoiding extensions of earlier ones; the count shows one is always available. ■

Theorem 9.7 (Noiseless coding theorem)

Every prefix code for X has expected length $\sum_i p_i n_i \ge H(X)$, and there is a prefix code with expected length $< H(X) + 1$.

Proof. With $q_i = 2^{-n_i} / \sum_k 2^{-n_k}$, as in Proposition 9.5(b), $-\sum p_i \log_2 p_i \le -\sum p_i \log_2 q_i = \sum p_i n_i + \log_2 \sum_k 2^{-n_k} \le \sum p_i n_i$ (Kraft). Upper bound: $n_i = \lceil \log_2(1/p_i) \rceil$ has $\sum 2^{-n_i} \le 1$ and $n_i < \log_2(1/p_i) + 1$. ■

Coding blocks of k copies gets within $1/k$ of $H(X)$ per symbol: entropy is the minimal bits per symbol.

Example 9.6

Probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$: $H(X) = 1.75$ bits, attained by $0, 10, 110, 111$; a fixed-length code needs 2.

Noisy channels. If each bit flips independently w.p. p, Shannon's theorem: rates below the **capacity** $C = 1 + p \log_2 p + (1-p) \log_2 (1-p) = 1 - H(\text{Bernoulli}(p))$ admit codes with error probability $\to 0$; rates above C do not.

9.5 Selected exercises

Motivation. Ross's problems and exercises (book numbering): a Poisson computation, a stationary law by symmetry, and entropy chain-rule consequences.

Exercise 9.1 (Two arrivals in the first hour)

Customers arrive at a bank as a Poisson process with rate λ. Given that two customers arrived during the first hour, find the probability that (a) both arrived during the first 20 minutes; (b) at least one did.

Solution.

Step 1. In hours, $N(1/3)$ and $N(1) - N(1/3)$ are independent Poisson$(\lambda/3)$, Poisson$(2\lambda/3)$, and $N(1) \sim$ Poisson(λ) (Proposition 9.1).

Step 2. For $k = 0, 1, 2$,

$$P(N(1/3) = k \mid N(1) = 2) = \frac{e^{-\lambda/3}\frac{(\lambda/3)^k}{k!}\, e^{-2\lambda/3}\frac{(2\lambda/3)^{2-k}}{(2-k)!}}{e^{-\lambda}\lambda^2/2} = \binom{2}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{2-k},$$

$\mathrm{Bin}(2, \frac{1}{3})$, free of λ (the arrivals act as independent uniform times).

(a) $P(N(1/3) = 2 \mid N(1) = 2) = 1/9$.

(b) $P(N(1/3) \geq 1 \mid N(1) = 2) = 1 - (2/3)^2 = 5/9$. ■

Exercise 9.7 (Doubly stochastic chains)

A transition matrix on $\{0, 1, \ldots, M\}$ is doubly stochastic if $\sum_{i=0}^{M} P_{ij} = 1$ for every j. Show that if such a chain is ergodic, then $\pi_j = 1/(M+1)$ for all j.

Solution.

Step 1. $\pi_j = 1/(M+1)$ has $\sum_j \pi_j = 1$ and

$$\sum_{i=0}^{M} \pi_i P_{ij} = \frac{1}{M+1}\sum_{i=0}^{M} P_{ij} = \frac{1}{M+1} = \pi_j.$$

Step 2. By uniqueness in Theorem 9.4, these are the limiting probabilities. ■

Exercise 9.17 (Functions do not increase entropy)

Show that $H(f(X)) \leq H(X)$ for every discrete X and every function f.

Solution.

Step 1. With $Y = f(X)$, $(X, Y) = (x, f(x))$ w.p. $P(X = x)$, so $H(X, Y) = H(X)$.

Step 2. $H(X, Y) = H(Y) + H_Y(X)$ (Proposition 9.5(a)) and $H_Y(X) \geq 0$.

Step 3. So $H(Y) \leq H(X)$, with equality iff $H_Y(X) = 0$, i.e. f is one-to-one on the values of X. ■

Exercise 9.18 (Capacity of the binary symmetric channel)

A bit X is sent and Y received, with $P(Y = 1 \mid X = 1) = P(Y = 0 \mid X = 0) = p$. The rate of transmission is $H(X) - H_Y(X)$ and the capacity is its maximum over $\alpha = P(X = 1)$. Show that the maximum is attained at $\alpha = \frac{1}{2}$ and equals $1 + p\log_2 p + (1-p)\log_2(1-p)$.

Solution.

Step 1. $H(X) + H_X(Y) = H(X, Y) = H(Y) + H_Y(X)$, so the rate is $H(Y) - H_X(Y)$.

Step 2. With $h(p) = -p\log_2 p - (1-p)\log_2(1-p)$, $H(Y \mid X = x) = h(p)$ for both x, so $H_X(Y) = h(p)$ for every α.

Step 3. $H(Y) \leq 1$, with equality iff $P(Y = 1) = \alpha p + (1-\alpha)(1-p) = \frac{1}{2}$, e.g. $\alpha = \frac{1}{2}$.

Step 4. So the capacity is $1 - h(p)$, the C of Section 9.4. ■

Lecture 10. Simulation

10.1 Random numbers and simulation

Intractable probabilities and expectations are estimated by **simulation**: average many independent realizations (strong law). Everything is built from **random numbers**, i.i.d. $U(0,1)$ (pseudorandom in practice); e.g. $\lfloor nU \rfloor + 1$ is uniform on $\{1, \dots, n\}$.

Example 10.1 (A random permutation)

Start from any ordering $P_1, \dots, P_n$; for $k = n, \dots, 2$ choose I uniform on $\{1, \dots, k\}$ and swap P_I, P_k. Position n gets each element w.p. $1/n$, then position $n-1$ each remaining one w.p. $1/(n-1)$, and so on: probability $1/n!$. The first r positions filled give a random r-subset (random assignment to treatments).

10.2 Simulating continuous random variables

Motivation. All distributions are built from uniforms, by inverse transforms or rejection.

10.2.1 The inverse transformation method

Proposition 10.1

If F is a continuous cdf and $U \sim U(0,1)$, then $X = F^{-1}(U)$ has cdf F.

Proof. $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$, since F is increasing. ∎

Example 10.2

(a) Exponential: $X = -\frac{1}{\lambda}\log(1-U)$, or $-\frac{1}{\lambda}\log U$.
(b) Gamma(n, λ), integer n: $X = -\frac{1}{\lambda}\log(U_1 U_2 \cdots U_n)$ (one logarithm).

10.2.2 The rejection method

Suppose we can simulate from g and $f(y)/g(y) \le c$ for all y.

Theorem 10.2 (Rejection method)

Repeat: generate $Y \sim g$ and an independent $U \sim U(0,1)$, until $U \le f(Y)/\{cg(Y)\}$; then set $X = Y$. The resulting X has density f, and the number of iterations is geometric with mean c.

Proof. In one iteration, $P(\text{accept}) = \int P(U \le \frac{f(y)}{cg(y)}) g(y)\,dy = \int \frac{f(y)}{c}\,dy = \frac{1}{c}$, and

$$P(Y \le x \mid \text{accept}) = c \int_{-\infty}^{x} \frac{f(y)}{cg(y)} g(y)\,dy = F(x).$$

Iterations are independent, so their number is geometric with parameter $1/c$. ∎

Example 10.3 (Normal from exponentials)

$|Z|$ has density $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$, $x > 0$. With $g(x) = e^{-x}$,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}}\, e^{x - x^2/2} \le \sqrt{\frac{2e}{\pi}} = c \approx 1.32,$$

attained at $x = 1$, and $\frac{f(x)}{cg(x)} = \exp\{-(x-1)^2/2\}$. Generate $Y = -\log U_1$, accept if $U_2 \le e^{-(Y-1)^2/2}$, then attach a random sign; about 1.32 iterations per normal.

Example 10.4 (The polar method)

By Example 6.14, $R\cos\Theta$, $R\sin\Theta$ are i.i.d. $N(0,1)$ for independent $R^2 \sim \mathrm{Exp}(1/2)$, $\Theta \sim U(0, 2\pi)$. Avoiding trigonometry: generate $V_i = 2U_i - 1$ until $S = V_1^2 + V_2^2 \le 1$; then S is uniform and independent of the angle, with $\cos\Theta = V_1/\sqrt{S}$, $\sin\Theta = V_2/\sqrt{S}$. With $R^2 = -2\log S$,

$$X = V_1\sqrt{\frac{-2\log S}{S}}, \qquad Y = V_2\sqrt{\frac{-2\log S}{S}}$$

are i.i.d. $N(0,1)$; the acceptance probability is $\pi/4$.

χ^2_n: $\sum Z_i^2$, or $-2\log(U_1 \cdots U_{n/2})$ for even n, plus one Z^2 for odd n.

10.3 Simulating from discrete distributions

For $P(X = x_j) = p_j$, set $X = x_j$ if $\sum_{i<j} p_i \le U < \sum_{i\le j} p_i$ (discrete inverse transform).

Example 10.5

(a) Geometric: $X = j$ iff $1 - q^{j-1} \le U < 1 - q^j$, i.e. $X = 1 + \lfloor \log U / \log q \rfloor$ (with $1 - U$ for U).
(b) Binomial: $X = \sum_{i=1}^n \mathbb{1}\{U_i < p\}$.
(c) Poisson(λ): counting events of a rate-λ process in $[0,1]$ with gaps $-\log U_i/\lambda$,

$$N = \max\Big\{n : \prod_{i=1}^{n} U_i \ge e^{-\lambda}\Big\}$$

(multiply random numbers until the product drops below $e^{-\lambda}$; subtract 1).

10.4 Variance reduction

Averaging n copies to estimate $\theta = E[X]$ has error $\sqrt{\mathrm{Var}(X)/n}$; reducing the variance is as good as increasing n.

10.4.1 Antithetic variables

For identically distributed X_1, X_2, $\mathrm{Var}(\frac{X_1+X_2}{2}) = \frac{1}{4}\{2\,\mathrm{Var}(X_1) + 2\,\mathrm{Cov}(X_1, X_2)\}$, smaller than for independent copies when $\mathrm{Cov} < 0$. For $X_1 = h(U_1, \ldots, U_m)$ with h monotone in each argument, $X_2 = h(1 - U_1, \ldots, 1 - U_m)$ has the same law and $\mathrm{Cov}(X_1, X_2) \le 0$ (Exercise 10.A for $m = 1$).

10.4.2 Variance reduction by conditioning

By Proposition 7.6, $\mathrm{Var}(E[X \mid Y]) \le \mathrm{Var}(X)$ with the same mean: if computable, average $E[X \mid Y]$ instead of X.

Example 10.6 (Estimating π)

V_1, V_2 i.i.d. $U(-1,1)$, $I = \mathbb{1}\{V_1^2 + V_2^2 \le 1\}$: $E[I] = \pi/4$, $\mathrm{Var}(I) \approx 0.1685$. Conditioning on V_1,

$$E[I \mid V_1] = P(V_2^2 \le 1 - V_1^2 \mid V_1) = \sqrt{1 - V_1^2},$$

so average $\sqrt{1-U^2}$: $\mathrm{Var} = \frac{2}{3} - \frac{\pi^2}{16} \approx 0.0498$, a 3.4-fold reduction. Antithetic pairs $\frac{1}{2}\{\sqrt{1-U^2} + \sqrt{1-(1-U)^2}\}$ do better: about 0.0069 per pair, against 0.0249 for two independent values.

10.4.3 Control variates

If Y is simulated alongside X with known μ_Y, $X + c(Y - \mu_Y)$ has mean θ and variance $\mathrm{Var}(X) + c^2 \mathrm{Var}(Y) + 2c\,\mathrm{Cov}(X,Y)$, minimized by

$$c^* = -\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}, \qquad \mathrm{Var}(X + c^*(Y - \mu_Y)) = \mathrm{Var}(X)(1 - \rho(X,Y)^2).$$

In practice c^* is estimated from the pairs. In Example 10.6, $Y = U^2$ ($E[Y] = 1/3$) controls $\sqrt{1-U^2}$.

10.5 Selected exercises

Motivation. Ross's Chapter 10 problems (book numbering) verify the recipes; Exercise 10.A justifies antithetic negative correlation.

Exercise 10.1 (Random permutations by insertion)

Set $P(1) = 1$. For $k = 2, \dots, n$, generate a random number U, let $J = \lfloor kU \rfloor + 1$, and set $P(k) = P(J)$ and then $P(J) = k$ (so for $J = k$ simply $P(k) = k$). Show that after step k, $(P(1), \dots, P(k))$ is a uniformly random permutation of $1, \dots, k$.

Solution.

Step 1. Step k puts k at position J and moves the old $P(J)$ to position k; J is uniform and independent of earlier steps.

Step 2. The map $\Phi : (\sigma, j) \mapsto \tau$ (σ a permutation of $1, \dots, k-1$, $j \le k$) is a bijection onto permutations of $1, \dots, k$: j is the position of k in τ, and σ is recovered by putting $\tau(k)$ back at j ($j < k$) and deleting position k.

Step 3. (Induction.) If each σ has probability $1/(k-1)!$, each pair (σ, j) has $\frac{1}{(k-1)!} \cdot \frac{1}{k}$, so each τ has $1/k!$. ■

Exercise 10.7 (The maximum of random numbers)

Let $F(x) = x^n$ for $0 < x < 1$.
(a) Simulate from F using a single random number.

(b) Show that $\max(U_1, \dots, U_n)$ has distribution F for independent random numbers $U_1, \dots, U_n$, and so give a second method.

Solution.

(a) $F^{-1}(u) = u^{1/n}$, so $X = U^{1/n}$ (Proposition 10.1): $P(U^{1/n} \le x) = P(U \le x^n) = x^n$.

(b) $P(\max_i U_i \le x) = \prod_i P(U_i \le x) = x^n$: a second (costlier) method. ■

Exercise 10.10 (Choosing the exponential envelope)

In Example 10.3, use the envelope $g_\lambda(x) = \lambda e^{-\lambda x}$, $x > 0$, instead of e^{-x}. Show that the mean number of iterations of the rejection method is minimized when $\lambda = 1$.

Solution.

Step 1. With $f(x) = \sqrt{2/\pi}\, e^{-x^2/2}$,

$$\frac{f(x)}{g_\lambda(x)} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda x - x^2/2} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda^2/2} e^{-(x-\lambda)^2/2},$$

maximal at $x = \lambda$, so the best constant is $c(\lambda) = \sqrt{2/\pi}\, e^{\lambda^2/2}/\lambda$.

Step 2. The mean number of iterations is $c(\lambda)$ (Theorem 10.2), and $\frac{d}{d\lambda} \log c(\lambda) = \lambda - \frac{1}{\lambda}$ changes sign at $\lambda = 1$: minimum $c(1) = \sqrt{2e/\pi} \approx 1.32$. ■

Exercise 10.13 (Polar coordinates of a uniform point in the disc)

Let (X, Y) be uniform on the unit disc, with density $1/\pi$ there, and let (R, Θ) be its polar coordinates, $\Theta \in (0, 2\pi)$.
(a) Show that R and Θ are independent, $R^2 \sim U(0,1)$ and $\Theta \sim U(0, 2\pi)$.
(b) Show that if V_1, V_2 are independent $U(-1, 1)$, then given $V_1^2 + V_2^2 \le 1$ the point (V_1, V_2) is uniform on the unit disc. (Together, (a) and (b) justify the polar method, Example 10.4.)

Solution.

(a)

Step 1. $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$ maps $(0,1) \times (0, 2\pi)$ one-to-one onto the disc minus a null segment, with Jacobian r; by Theorem 6.3

$$f_{R,\Theta}(r, \theta) = \frac{1}{\pi} \cdot r = 2r \cdot \frac{1}{2\pi}, \qquad 0 < r < 1,\ 0 < \theta < 2\pi.$$

Step 2. It factors (Proposition 6.1), and $P(R^2 \le s) = \int_0^{\sqrt{s}} 2r\, dr = s$.

(b) (V_1, V_2) is uniform on $[-1, 1]^2$ (area 4), the disc D has area π; for $A \subseteq D$

$$P((V_1, V_2) \in A \mid (V_1, V_2) \in D) = \frac{\text{area}(A)/4}{\pi/4} = \frac{\text{area}(A)}{\pi},$$

■

Exercise 10.A (Antithetic variables are negatively correlated)

Let h be nondecreasing on $(0,1)$ with $E[h(U)^2] < \infty$ for $U \sim U(0,1)$. Show that $\text{Cov}(h(U), h(1-U)) \le 0$.

Solution. [Supplement]

Step 1. $k(u) = h(1-u)$ is nonincreasing, so for all u, v

$$\big(h(u) - h(v)\big)\big(k(u) - k(v)\big) \le 0.$$

Step 2. With $V \sim U(0,1)$ independent of U, take expectations and expand (U, V i.i.d.):

$$0 \ge E[h(U)k(U)] - E[h(U)]E[k(V)] - E[h(V)]E[k(U)] + E[h(V)k(V)] = 2\,\mathrm{Cov}\big(h(U), k(U)\big).$$

(finite by Cauchy–Schwarz, as $k(U) \overset{d}{=} h(U)$).

Step 3. So the antithetic estimator has variance at most $\frac{1}{2}\,\mathrm{Var}(h(U))$, that of two independent copies. ■